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Web API Specification
for iPECS SME System

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Version history

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0.9	2018.08.17	Initial Release SMDR and click to call feature added for SME system.
0.99	2020.01.07	Web API Document Release SMDR, Click to Call, ACD and Call Control API
1.0A	2020.02.25	Trouble Shooting added Conference Room Command added callRefID added in SMDR Direction Code added in SMDR
1.0B	2020.04.23	Clear connection resource name was changed from clear_connection to clear in this document. Reject call was separated from clear connection.
1.0C	2020.06.15	SMDR data modified. SMDR menu added in apidemo page
1.0D	2020.09.17	ACD data modified. (TIME & DATE) Station Name added in SMDR data
1.0E	2021.05.25	Correction of errata
1.0F	2021.11.10	Get monitor status added Get presence status added CO check API added

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1. Overview 1
2. Getting Started 2
2.1. Configuration (admin program) 2
2.1.1. Set Up Account 2
2.1.2. Run Web API service 3
2.2. API Demo Page (System built-in) 4
2.2.1. Access Demo Page 4
2.2.2. Page Layout Description 5
2.3. Trouble Shooting 6
2.3.1. Suspicious user handling 6
3. Access Web API 8
3.1. API URI 8
3.2. Authentication scheme 9
3.2.1. Login 9
3.2.2. Check Connection Request 11
3.2.3. Logout 11
4. GET query rule 15
4.1. Query with JSON data 15
4.2. Regular expression 15
5. API Resource 17
5.1. SMDR 17
5.1.1. Examples with curl utility application 21
5.1.2. Service Example 22
5.2. ACD 23
5.3. CTI (Call Control) 37
5.3.1. Term Description 37

5.3.2. Call Event Start and Stop Request 38
5.3.2.1. Monitor Start 38
5.3.2.2. Monitor Stop 39
5.3.2.3. Get Monitor Status 40
5.3.2.4. Call Event Message 42
5.3.2.4.1. Dialing Event 43
5.3.2.4.2. Proceeding Event 43
5.3.2.4.3. Alerting Event 44
5.3.2.4.4. Ringing Event 44
5.3.2.4.5. Connected Event 45
5.3.2.4.6. Released Event 46
5.3.2.4.7. ReleasedAsc Event 46
5.3.2.4.8. Rerouted Event 47
5.3.2.4.9. Rerouting Event 47
5.3.2.4.10. Failed Event 48
5.3.2.4.11. Held Event 49
5.3.2.4.12. Hold Event 49
5.3.2.4.13. Retrieved Event 50
5.3.2.4.14. Retrieve Event 50
5.3.2.4.15. Transferred Event 51
5.3.2.4.16. Transferring Event 52
5.3.2.4.17. Conferenced Event 52
5.3.2.4.18. Conferencing Event 53
5.3.2.4.19. DoNotDisturb Event 53
5.3.2.4.20. Forwarding Event 54
5.3.2.4.21. Registered Event 55
5.3.2.4.22. UnRegistered Event 55
5.3.2.4.23. MWI Event 55
5.3.2.4.24. ACD Agent LogOn Event 56
5.3.2.4.25. ACD Agent LogOff Event 57
5.3.2.4.26. ACD Agent Ready Event 57
5.3.2.4.27. ACD Agent Not Ready Event 58
5.3.2.4.28. ACD Agent Work Event 59
5.3.2.4.29. ACD Agent Busy Event 59
5.3.2.4.30. ACD Group Queueing Event 60
5.3.2.4.31. ACD Group Queued Event 61
5.3.2.4.32. ConfGroup Member Event 61
5.3.2.4.33. ConfGroup muteOn Event 62

5.3.2.4.34. ConfRoom Member Event 62
5.3.2.4.35. ConfRoom Usage Event 63
5.3.3. Presence Event Start and Stop 65
5.3.3.1. Presence Start 65
5.3.3.2. Presence Stop 66
5.3.3.3. Get Presence Status 68
5.3.3.4. Presence Event 69
5.3.4. CO Status Event Start and Stop 70
5.3.4.1. Check CO Start 70
5.3.4.2. Check CO Stop 72
5.3.4.3. Get Check CO Status 73
5.3.4.4. Check CO Event 75
5.3.5. Call Control Command 77
5.3.5.1. Get Terminal Information 77
5.3.5.2. Make Call 78
5.3.5.3. Answer Call 80
5.3.5.4. Clear Connection 81
5.3.5.5. Reroute Call 82
5.3.5.6. Hold Call 84
5.3.5.7. Retrieve Call 86
5.3.5.8. Consultation Call 87
5.3.5.9. Transfer Call 89
5.3.5.10. Sigle Step Transfer Call 91
5.3.5.11. Generate Digit(DTMF) 92
5.3.5.12. Reconnect Call 94
5.3.5.13. Set Forwarding 95
5.3.5.14. Get Forwarding 96
5.3.5.15. Set DND 98
5.3.5.16. Get DND 99
5.3.5.17. PickUp 100
5.3.5.18. Conference Call 102
5.3.5.19. Call Reject 103
5.3.5.20. BusyService 105
5.3.5.21. Set Call Back 106
5.3.5.22. Get Call Status 107
5.3.5.23. Call Park 109
5.3.5.24. Call UnPark 110
5.3.5.25. Get MWI (Message Waiting Indicator) 111

5.3.5.26. Flexible Page 113
5.3.5.27. ACD Agent LogOn 114
5.3.5.28. ACD Agent LogOff 115
5.3.5.29. ACD Agent Set State 117
5.3.5.30. ACD Group Control 118
5.3.5.31. Conference Group Call Start 118
5.3.5.32. Conference Group Call Add Callee 120
5.3.5.33. Conference Group Call Drop Callee 121
5.3.5.34. Conference Group Call Set Mute On 122
5.3.5.35. Get conference room state 124
5.3.5.36. Conference Room Open/Close 125
5.4. Click to Call 128
6. Error Code Table 129

1. Overview

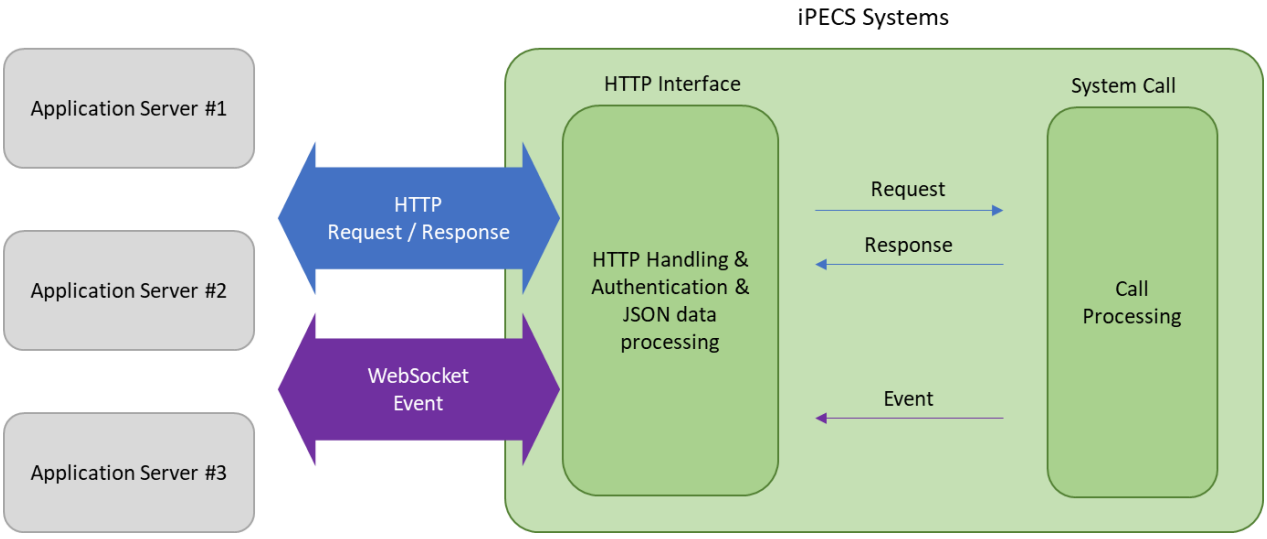
iPECS SME System Web API is a platform independent API that can directly control iPECS SME system with HTTP based API. This API helps you build applications to provide more services to your customers. You can use APIs to connect to our system after authentication to control or monitor some of its resources. This API provides the following features:

- SMDR API: providing SMDR information after every call completion.
- ACD Manager Protocol API: providing all call event information in a real time.
- Call Control and Event API: providing call control service to control and monitor CTI functionality in the system.
- ClickCall API: providing a web service to make a call.

License is required to use each group of Web APIs as below.

- Call Control and Call Event APIs: Server license (API-CTI) and Client license (API-CTI-EXT) as the number of clients if there are CTI clients to use call control APIs.
- SMDR APIs: API-SMDR
- ACD Manager Protocol APIs: API-AMP
- ClickCall API: CLICKCALL license as the number of clients.

Because it works over HTTPS, the connection with the server is encrypted. Requests and responses are served over HTTP and you can receive information raised as events when you connect a WebSocket (Server Type Only). All data is provided in JSON format. You can use multiple functions at the same time through one connection with one login. And multiple logins with the same account are not allowed. Up to three Web API accounts can be logged in.

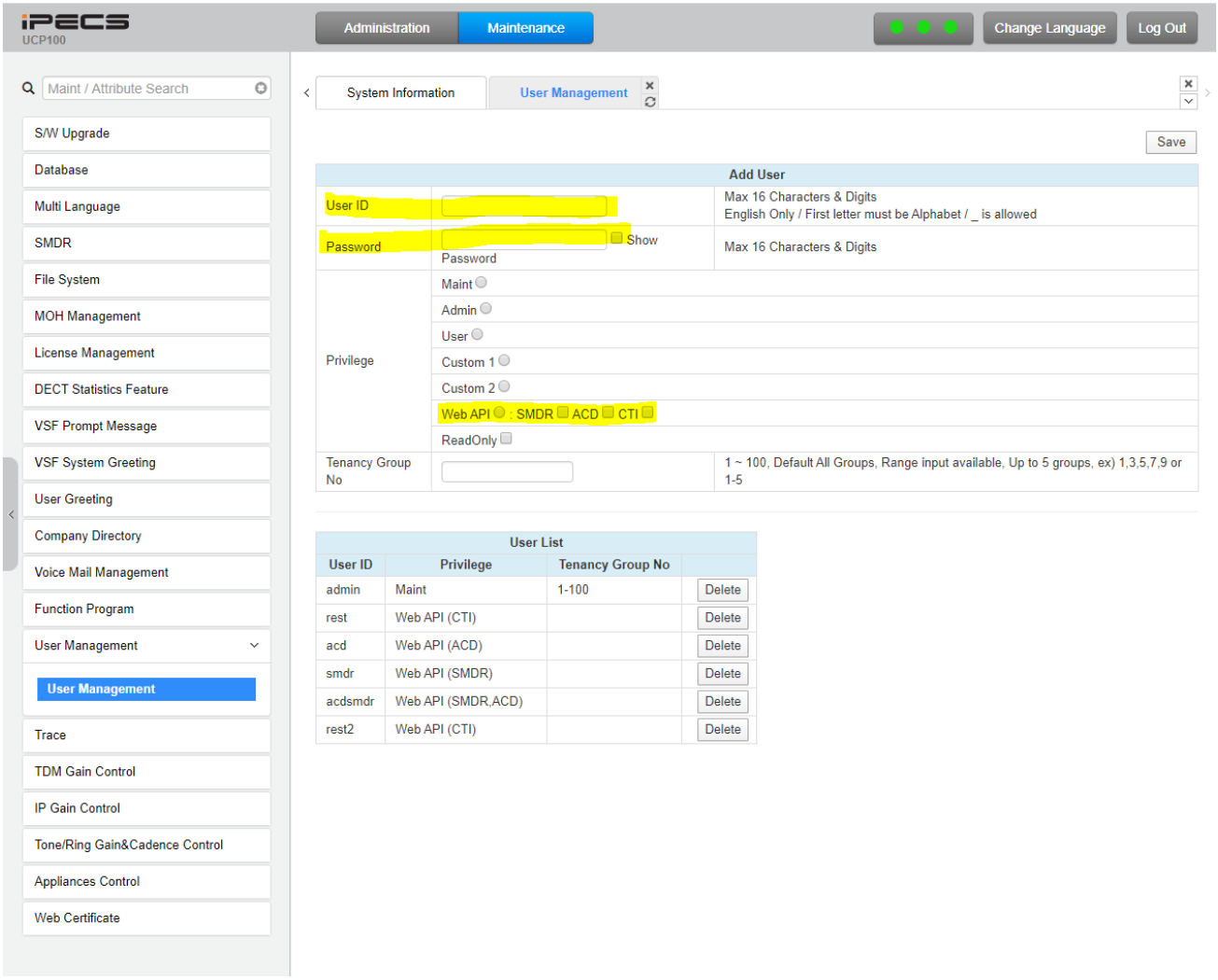


2. Getting Started

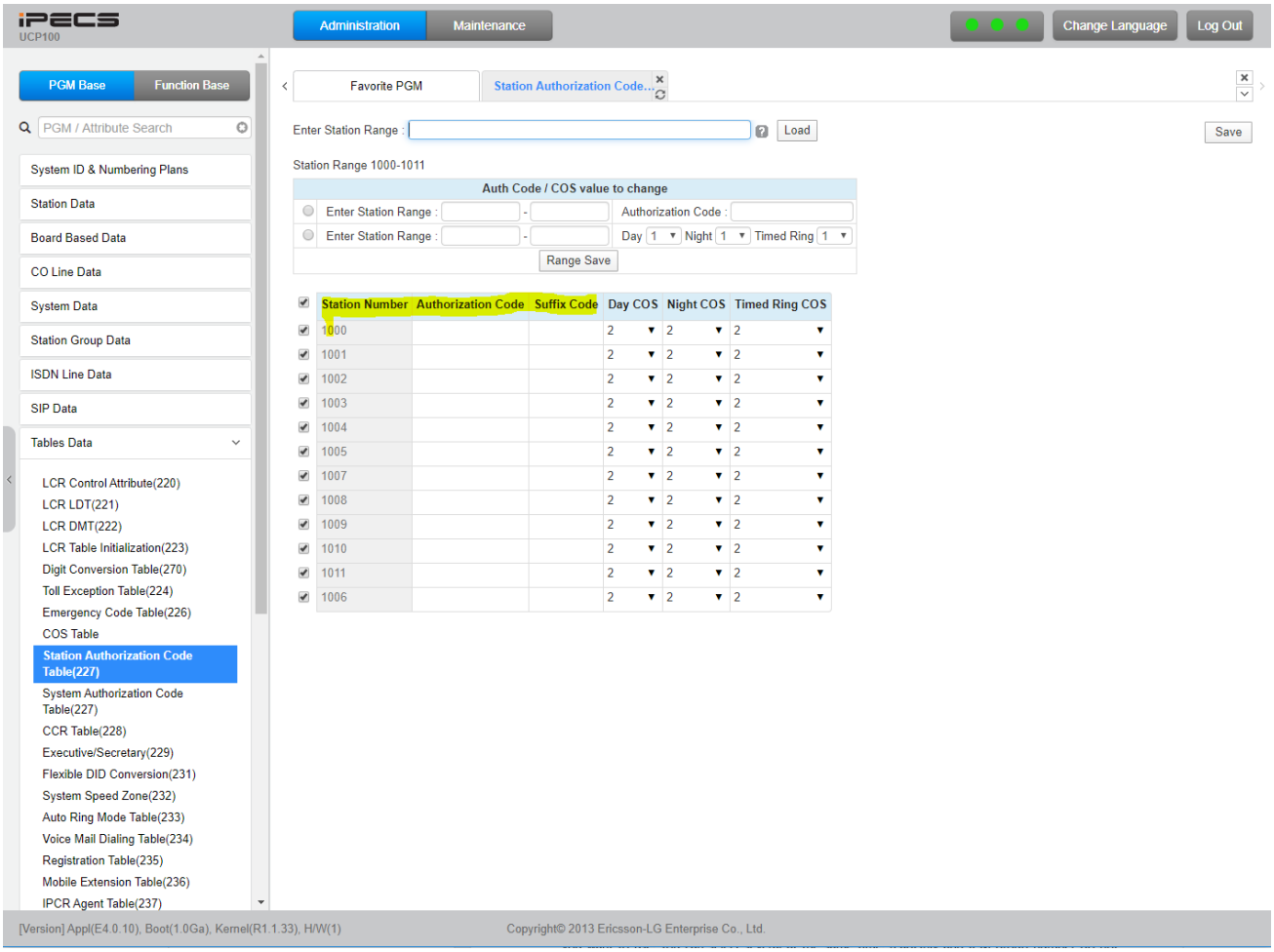
2.1. Configuration (admin program)

2.1.1. Set Up Account

You need to create an account to use the Web API (Sever Type). This account is for the server and has all the permissions provided by the API. Please contact your system administrator. System administrator can access Web Admin and go to Maintenance->User Management->User Management page. Enter the proper ID and password, select Web API in the Privilege section, and check the features you want to use. You can select several resources (SMDR, ACD or CTI) and you can use resources at once with one login. readOnly and ICM group number do not matter.

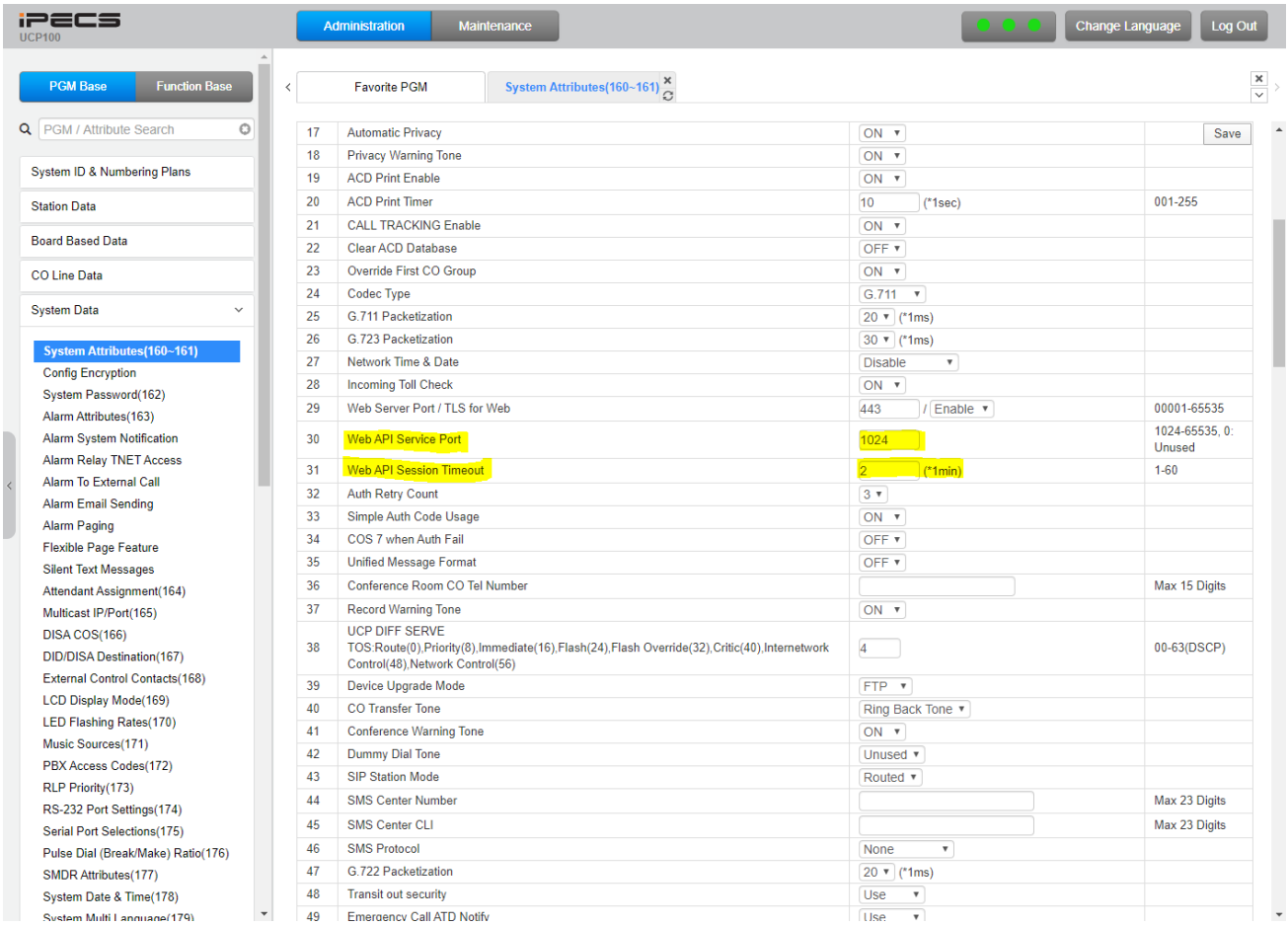


The Station user uses the station number as the ID and the password as a combination of the Authentication Code and the Suffix Code in the Administration->Table Data->Station Authentication Code Table(227). For example, if the Authentication Code is *1234 and the Suffix Code is Secr01$, the password you must enter to log in is *1234Secr01$. We recommend using a combination of the two code for security reasons. If only one of them is programmed, the other can be omitted. If neither is programmed, login is restricted.



2.1.2.Run Web API service

You must start the service to use the web API. Please contact your system administrator. System administrator connects to Web Admin and goes to main program program-> System related data-> System characteristic (160 ~ 161) page. Specifies the web API service port as the port not used by the system from 1024 to 65535. Specifies the web API session duration. The initial value is 2 minutes. This is the time to maintain authentication after closing the connection. 2 minutes after the client makes the last request, the token expires, and the request is rejected. However, if the WebSocket is connected, authentication continues.



2.2. API Demo Page (System built-in)

If you install a version that supports the iPECS SME system Web API (4.0 or later, check the detailed version), you can use the API Demo Page. You can try out the API before developing it or debug it during development. Only provide CTI API requests, and events can be used together depending on your account's permissions.

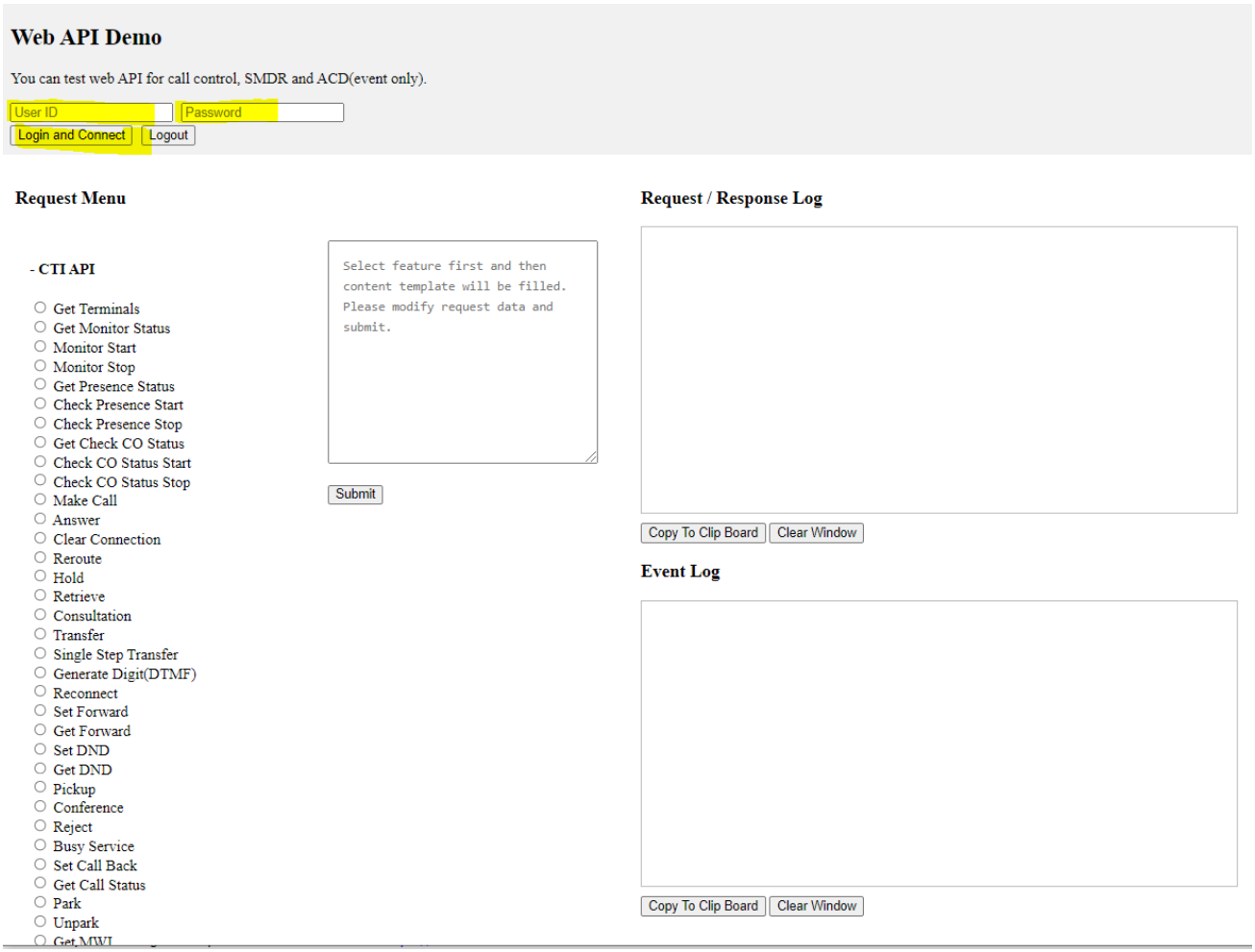
2.2.1. Access Demo Page

After starting the Web API service, use a web browser to connect to the following URI.

https://{IP ADDRESS}:{PORT NUMBER}/apidemo.html

* {PORT NUMBER} : Web Server Port, Not Web API Service Port

When connected, the following screen appears. This page is a simple demo or test page for application development.



2.2.2. Page Layout Description

After entering ID (Server Type Only) and Password, click “Login & Connect” button to login and WebSocket will try to connect.

If you select Request Menu, the template to be used in each request is displayed in the edit box. After modifying the contents, click the Submit button to send the request.

Request Log is recorded in Request / Response Log box. You can easily copy the contents by pressing the “Copy To Clip Board” button, and erase them all by pressing the “Clear Window” button.

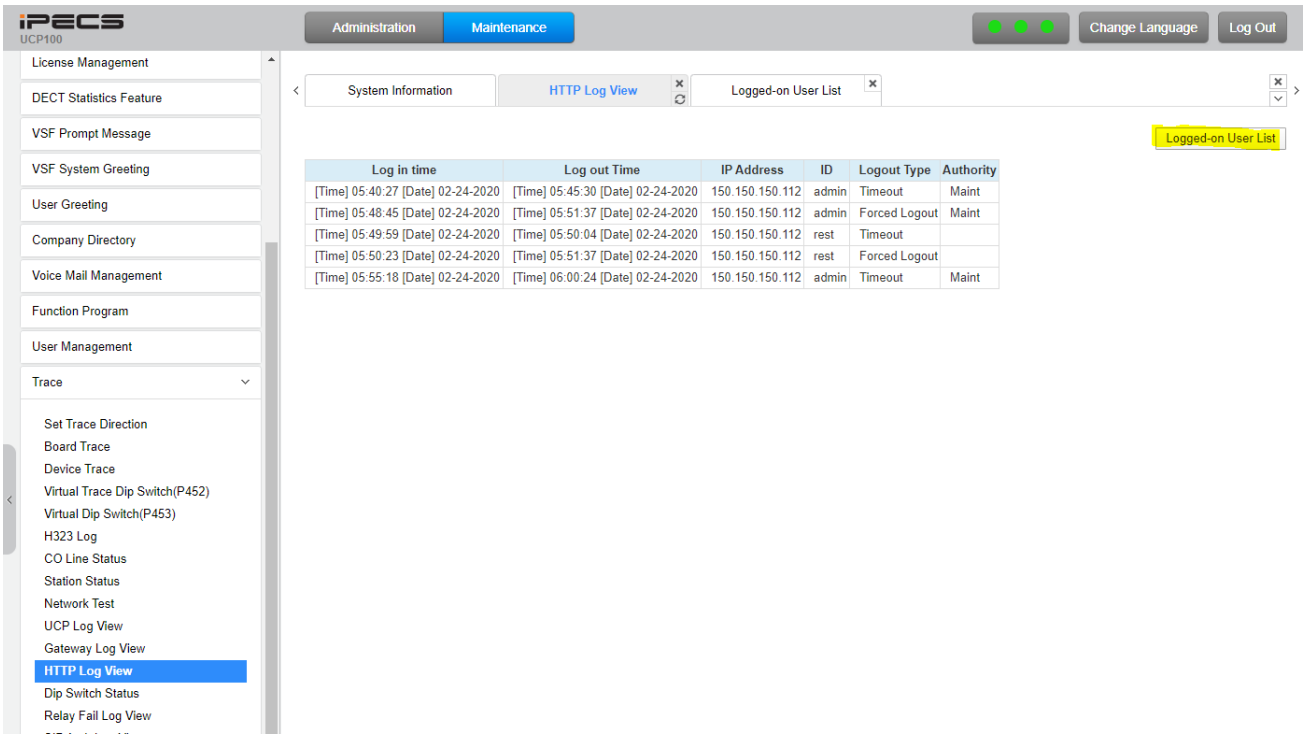
Event data is output to the Event Log box. The functions of the buttons below are the same as the Request / Response Log boxes.

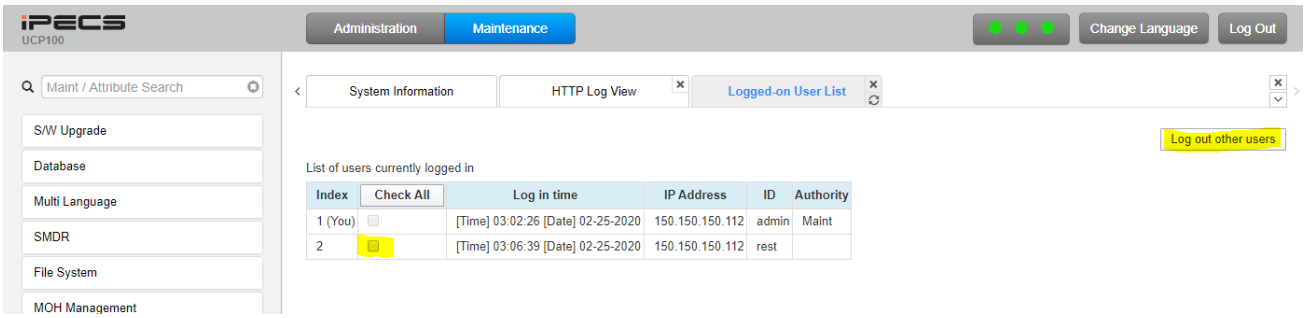
For low level debugging of HTTP, you can use the inspection function provided by web browser.

2.3. Trouble Shooting

2.3.1. Suspicious user handling

If you get an "already login" message when you try to login, the web server may be cleaning up the session, so try again a few seconds later. If it's the same after a few seconds, check Web API sessions at the Web Admin page(Maintenance->Trace->HTTP Log View->Logged-on User List) if you or someone have logged in elsewhere.





If an unknown user is logged in, you can logout that user. Check the user's checkbox and press the "Log out other users" button.

3. Access Web API

Basic information for accessing to the Web API server.

3.1. API URI

(Application Program Interface Uniform Resource Identifier, similar as URL)

{BASE URI} : https://{IP ADDRESS}:{PORT NUMBER}/ipxapi

The base URI consists of IP address and TCP port number for API server and ipxapi string. The API uses HTTPS only for security and ipxapi means iPECS REST API resource. We recommend using certificate issued by Trusted Root Certification Authority or self-signed on your web server (system) for security enhancement when HTTPS session is established.

{FULL URI} : {BASE URI}/{TYPE}/{VERSION}/users/{USER ID}/{RESOURCE}/{FEATURE}

The full URI for service consists of as above.

{TYPE} means resource type for server or client.
{VERSION} means API version to access.
users means user ID will be followed next.
{USER ID} means user ID.
{RESOURCE} means resource name to access.
{FEATURE} means feature of resource. Now CTI uses only.

Example) https://10.10.10.1:1024/ipxapi/server/v1/users/test/login
The example means user ID test tries to login to access and use server type resource.

There is a resource that provides event data when a WebSocket is connected. E.g. SMDR, ACD and CTI. After logging in with an account that has the authority of the relevant resource, if you connect only one WebSocket, you can receive all event data.

{WebSocket URI} : wss://{IP ADDRESS}:{PORT NUMBER}/ipxapi/{TYPE}/{VERSION}/users/{User ID}

3.2. Authentication scheme

To access Web API, you can login using the ID and password via HTTP basic authentication (RFC 7235) to get authentication token. If success, you will get token string.

3.2.1. Login

Feature	Login
Description	Get authentication token
Service Type	Server/Client
Method	GET
URI	{BASE URI}/{server | client}/{VERSION}/users/{User ID}/login
Request Data	N/A, use HTTP header, Basic Authentication (RFC 7617) - Header example Accept: */* Accept-Encoding: gzip, deflate, br Accept-Language: ko,en-US;q=0.9,en;q=0.8,he;q=0.7 Authorization: Basic cmVzdDphcGkxMDI0Nzc3 Cache-Control: no-cache Connection: keep-alive Content-type: application/json Cookie: SessionID=14864678+0 DNT: 1 Host: 150.150.150.113:1024 Pragma: no-cache Referer: https://150.150.150.113:1024/apidemo.html Sec-Fetch-Mode: cors Sec-Fetch-Site: same-origin User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/79.0.3945.117 Safari/537.36
Return Format	Normal case 200 OK in header { "token": "{Token from system}", "checkConnectionTimer": {second},

	"systemType": "iPECS [XXX]",
	“systemVersion”: “{VERSION}”,
	"apiVersion": "1.0.0"
	}
	
	Error case
	{
	"error": {
	"code": ###,
	"message": "#Message"
	}
	}

In case of Positive Response, information about connected system and interface specifications is transmitted.

[Common Attributes]

Parameter	Value	Mandatory/ Optional	Description
token	String	M	Token for Command Request API
feature	String	M	List the resources available for the logged in account, ex) ACD,SMDR,CTI
checkConnectionTimer	Positive integer	M	Session Timeout after login. You must update the timer within this time to keep the login. Refer Check Connection Request.
systemType	String	M	System Type, ex) iPECS UCP100
systemVersion	String	M	System Version, ex) S-UCP-Rls2001-4.0.11-App
apiVersion	String	M	Web API version, ex) 1.0.0

[Specific Attributes]

Parameter	Value	Resource	Mandatory/ Optional	Description
terminalCount	Positive integer	CTI	M	Terminal total count

In case of Negative Response, return Error value as below.

- Operation (serviceNotSupported) - service not provided

3.2.2.Check Connection Request

Web Client sends periodically Check Connection Request to maintain the session which is set by Login Request.

URI	{BASE URI}/{server | client}/{VERSION}/users/{User ID}/checkconnection
Method	**POST**
Header	Authorization: Bearer <token from login response>
Parameter	

Positive Response	HTTP 200 OK
	Content:
	{
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

[Condition]

• Web Client must update the session timer by sending a request within checkConnectionTimer value time If nothing else is sent.

• Web Client re-executes the Login Request authentication procedure when reconnecting after disconnecting the session.

• If the WebSocket is connected, Web Client can keep session using Ping via WebSocket instead of this request.

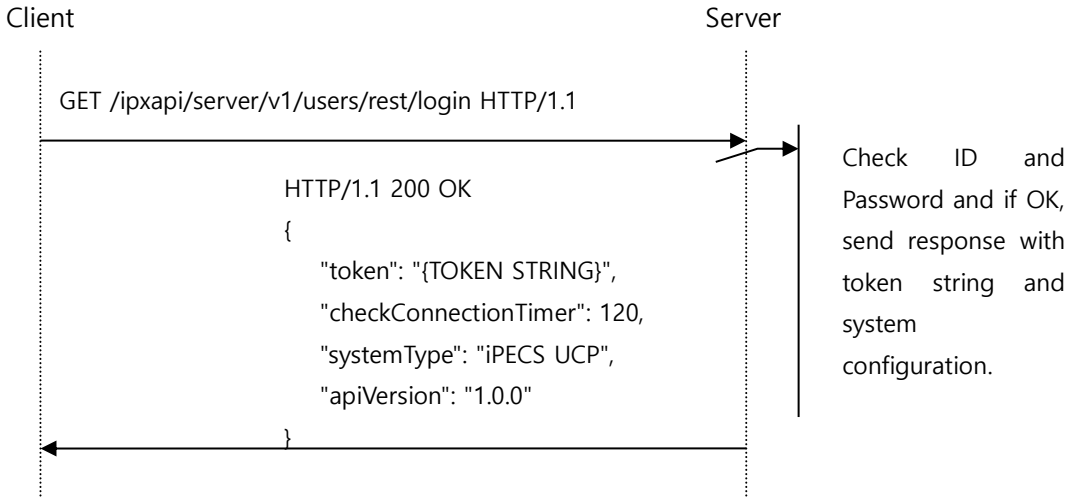
3.2.3.Logout

Feature	Logout
Description	Logout
Service Type	Server/Client
Method	POST
URI	{BASE URI}/{server | client}/{VERSION}/users/{User ID}/logout
Header element	Authorization : bearer {token from system}

Request Data	N/A

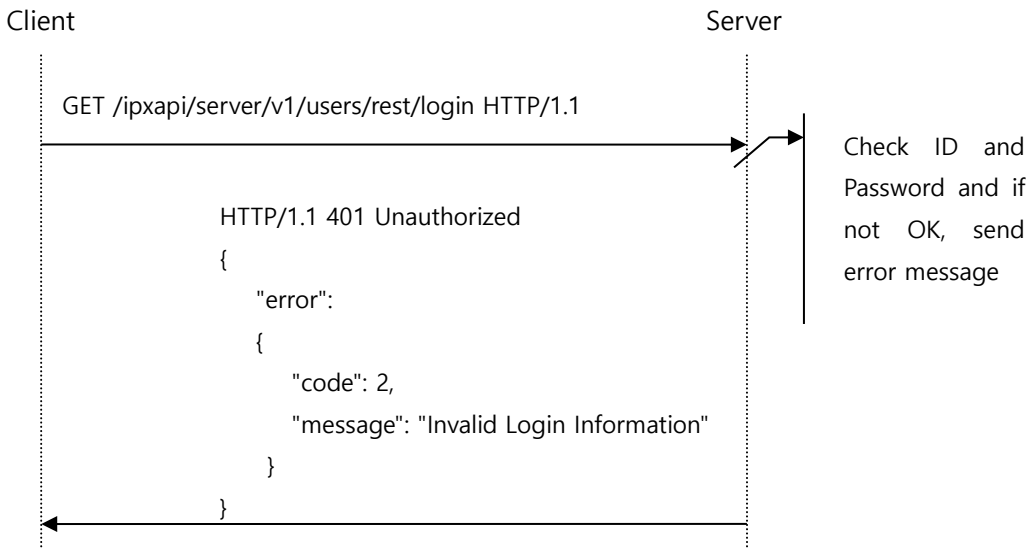
Return Format	Normal case
	200 OK in header without Content
	
	Error case
	{
	"error": {
	"code": ###,
	"message": "#Message"
	}
	}

Case 1) Login OK



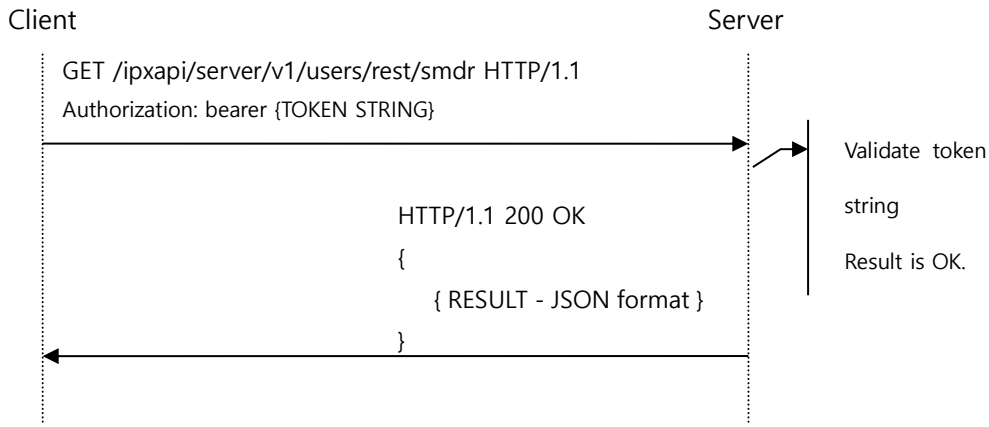
* The {TOKEN STRING} is JWT (JSON Web Token, rfc7519) string.

Case 2) Login Failure

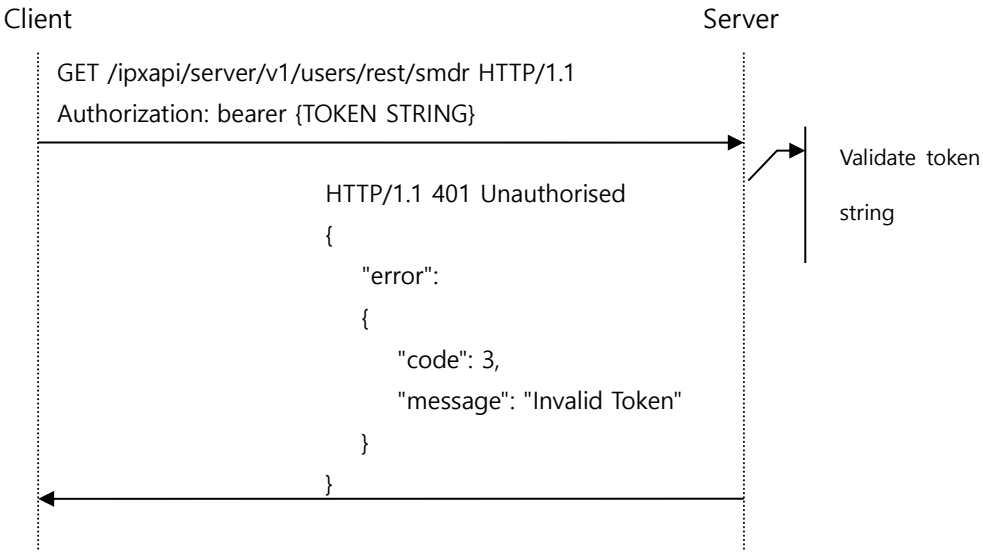


If you receive HTTP response code 401 Unauthorized, please try to login again.
If you fail again, please ask the system administrator.

If you succeed to get token, you can use it to access the system resource. The token should be added like Authorization: bearer {TOKEN STRING} in HTTP header. The server check validation of the token and then response result. If the result is OK, the server will process your request and response result.



If token validation is failed in server, the server will response with error code and message. The error code is explained in error code table.


Client
Server
GET /ipxapi/server/v1/users/rest/smdr HTTP/1.1
Authorization: bearer {TOKEN STRING}
Validate token string
HTTP/1.1 401 Unauthorised
{
"error":
{
"code": 3,
"message": "Invalid Token"
}
}

4. GET query rule

GET method means request to retrieve resource. To classify item from resource, you can (have to) use query string after {FULL URI}.

4.1. Query with JSON data

{FULL URI}?{QUERY}

The query string consists of name-value pairs according to HTTP1.1 standard as follow.

{QUERY} : {NAME#1}={VALUE#1}&{NAME#2}={NAME#2}& ... &q={--JSON DATA--}

The name-value pair is used. If you want to use query in json format, use q={--JSON DATA--} string. For example, if you want to request count of record when you use SMDR resource, use as q={"countOnly":true}. This rule is used throughout the API.

4.2. Regular expression

Regular expressions are available in SMDR resource. SMDR data can be filtered by regular expressions.

Basic query format: { "$operator" : { "left operand" : "right operand" } }

Operator name started from $ character in string. The operator is placed in string of JSON object and left operand and right operand pair is placed in object. The supported operators and examples are as below.

Operator	**Description**	**Example**
	Match	{"field" : value}
$not	Not match	{"$not" : {"field" : value}}
$or	Or operator	{"$or": [{"field" : value1}, {"field" : value2}, {"field2" : value3} ...]}
$and	And operator	{"$and": [{"field" : value1}, {"field2" : value2} ...]}
$regex	Match field	{"$regex" : {"field" : --FORMAT--}}
$>	Greater	{"$>" : {"field": 100}}

$>=	Same or greater	{"$>=" : {"field": 100}}
$<	Little	{"$<" : {"field": 100}}
$<=	Same or little	{"$<=" : {"field": 100}}

The regular expression operator is “$regex”. There are two wildcards used in conjunction with –FORMAT-- for regular expression.

* : The asterisk indicates zero or more occurrences of the preceding element.

? : Matches any single character

To match these wildcards by character, use \ first.

Description	format
Finds any values that start with string	“string*”
Finds any values that have string in any position	“*string*”
Finds any values that end with string	“*string”
Finds any values in six length string that starts with s and ends with g	“s????g”
Finds any values that have wildcard characters string like “***” in any position	“***”

* Date format is YYYY-MM-DD hh:mm:ss.

Example)

JSON	Meaning
{ “status” : “normal”, “age” : “20” }	status=normal and age=20
{ "$or": [{ "status": "normal", "age": 20} , { "status": "idle", "age": 30 }] }	(status=normal and age=20) or (status=idle and age=30)
{ "code" : "XYZ", "$or": [{ "status": "normal", "age": 20} , { "status": "idle", "age": 30 }] }	code=XYZ and ((status=normal and age=20) or (status=idle and age=30))
{ "status":"normal", "$>=": {"age": 20}, "$<=": {“age”: 30}, "$not": {“age”:25} }	(status=normal and age>=20 and age<=30 and age != 25)
{ “$regex” : {“status”: “*L”}	Find status value that end with ‘L’
{ “$regex” : {“status”: “n????l”}	Find status value that start with ‘n’ and follow any four characters and end with ‘l’

5. API Resource

5.1. SMDR

"SMDR Web API Service" must be set to On in System Data-> SMDR Attributes (117). The output may vary depending on the SMDR Attributes menu setting. Check the SMDR menu. API License – SMDR license is required. Please refer to the manual for other SMDR options and Meaning of content in the detailed description table below. The server account and the station user can use it, but the server account can read and delete data, and the station user can only read their own data. The account for the server can receive SMDR data in real time when the WebSocket is connected.

Feature	SMDR
Description	Get SMDR data and Delete SMDR data
Service Type	Server/Client
Method	GET, DELETE (Server only)
URI	Request: {BASE URI}/{server | client}/{VERSION}/users/{User ID}/smdr Event: wss://{IP ADDR}:{PORT}/ipxapi/server/v1/users/{User ID}
Header element	Authorization: bearer {token from system}
Request Data	GET countOnly : request only record count, true or false maxCount : max number of records that can be retrieved at one time, positive integer type, if maxCount value is not requested, default value is 50. Maximum value is 100. {QUERY} : can use fields in items as below return format. Refer example as below. DELETE {QUERY} : can use fields in items as below return format. Refer example as below.
Return Format	Normal case 200 OK in header { "count": ## "items": [{ "Index": "#",

	“callRefID”: ###,
	"StationNumber": "####",
	“StationName”: “####”,
	"CoNumber": "##",
	"Duration": "hh:mm:ss",
	"StartTime": "YYYY-MM-DD hh:mm",
	"Direction": "#",
	"*DialedNumber1*": "####",
	"Cost": "##",
	"AccountCode": "####",
	“*DialedNumber2*”: “####”,
	"DisconnectCause": "#CODE-#REASON"
	},
	…
	]
	}
	
	Error case
	{
	"error": {
	"code": ###,
	"message": "#Message"
	}
	}

Event Data Format	{ "event":"smdr", "data": { "Index":"7", "callRefID":4, "StationNumber":"2008", “StationName”:”MyStation”, "CoNumber":"0", "Duration":"00:00:04", "StartTime":"2020-06-16 14:51", "Direction":"E", "Dialed Number":"006 ", "Cost":"",

	"AccountCode":"", "CLI Number":"", "DisconnectCause":"4F" } }
Meaning of content	count: number of items items: array include each record Index: index of record in SMDR DB (If all records are deleted, the Index starts again from 1) callRefID: Call Tracking Number if set "CALL TRACKING Enable" in System Attributes(160~161) StationNumber: Source number, if called from CO, this field will be displayed COxxx. StationName: Station Name(programmed in PGM Station Name Display) is printed if set "Station Name Print (Web API Service only)" in PGM SMDR Attributes(177). CoNumber: CO line number Duration: Call time serviced StartTime: Call start time, YYYY-MM-DD hh:mm Direction: See Table 1. Direction Code *DialedNumber1*: CLI/Dialed/CPN number field. This field is not fixed name. It will be controlled by the SMDR Ring/CLI/CPN Service-I value in PGM 177. If the value is set CLI, this name will be displayed "CLI Number". In other cases, "Dialed Number" or "CPN Number" Cost: Call cost AccountCode: Account Code entered or Entry number of Authorization Code(Password) - STA-P[12341000] : Station 12341000 Authorization Code - SYS-P[1111] : System Auth Table 1111 Authorization Code. *DialedNumber2*: CLI/Dialed/CPN number field. Same as *DialedNumber1* except that it is not required. It will be controlled by the SMDR Ring/CLI/CPN Service-II value in PGM 177. DisconnectCause: Disconnect cause(reason) if there is cause. It will be controlled by SMDR Disconnect Cause value in PGM 177. Refer Disconnect Cause Code Table.

Direction Code	Description
t	External incoming call transfer
I	External incoming call
O	External outgoing call
T	External outgoing call transfer
H	External lost call during held
R	External lost call during ringing
G	External lost call during group call ringing/queued/announcement
E	Internal(ICM) make call
S	Internal(ICM) send SMS
P	Internal(ICM) make page
o	External outgoing unanswered
X	Hotel minibar call
V	External incoming call answered by VSF voice mail, when VSF Voice mail indication is set to "V (new indication)" in SMDR Attributes(177) menu
N	Networking outgoing call
n	Networking incoming call

Table 1. Direction Code

Disconnect Cause Code	Description
1	UnAllocated Number
3	No Route to Dest
8	Warning Tone
16	Normal Call Clearing
18	No User Responding
19	User No Answer
20	subscriber absent
21	Call Rejected
22	Number Changed
22	User DND
27	Destination Out of Order
28	Invalid Number format
30	Response to STATUS ENQUIRY
31	Normal, Unspecified
34	No Channel Available
38	Network Out Of Order
41	Temporary Failure
42	switching equipment congestion

44	Request Channel Unavailable
47	Resource Unavailable
50	Request Facility Not Subscribed
57	Bearer Capability No Authorized
58	Bearer capability Not Available
63	Service or option not available
65	Bearer capability Not Implemented
66	Channel Type Not Implemented
69	Request Facility Not Implemented
70	Only restricted digital information bearer capability is available
79	Service or option not implemented, unspecified
81	Invalid Call Reference
83	Call ID No exist
84	Call ID In Use
85	No Call Suspended
86	Call ID Cleared
88	Incompatible Destination
91	Invalid transit network selection
97	Message Type No Exist
98	Message Type Not Compatible with Call state
100	Invalid Information Element
102	timer expire
110	message with unrecognized parameter, discarded : Only Russia ISUP
111	Protocol Error
112	SIP Enblock Address Incomplete (LG Only)

Table 2. Disconnect Cause Code

5.1.1. Examples with curl utility application

1) Get count of all records

   curl -G https://{IP address}:{port}/ipxapi/server/v1/users/{ID}/smdr -k -v -H "Authorization: bearer $variable" --data-urlencode 'countOnly=true'

2) Get up to 10 from first record

   curl -G https://{IP address}:{port}/ipxapi/server/v1/users/{ID}/smdr -k -v -H "Authorization: bearer $variable" --data-urlencode 'maxCount=10'

3) Get data with StationNumber 1003

   curl -G https://{IP address}:{port}/ipxapi/server/v1/users/smdr/smdr -v -k -H "Authorization: bearer $variable" --data-urlencode "q={\"StationNumber\":\"1003\"}"

4) Get data with StationNumber ending in 3 and up to 10 records

curl -G https://{IP address}:{port}/ipxapi/server/v1/users/{ID}/smdr -k -H "Authorization: bearer $variable" --data-urlencode "q={\"\$regex\":{\"StationNumber\": \"*3\"}}" --data-urlencode 'maxCount=10'

5) Get data with StationNumber is 1003 and StartTime is since 2018-08-10

curl -G https://{IP address}:{port}/ipxapi/server/v1/users/{ID}/smdr -v -k -H "Authorization: bearer $variable" --data-urlencode "q={\"StationNumber\":\"1003\", \"$>\":{\"StartTime\": \"2018-08-10\"}}"

6) Delete index 1, 3 and 5

curl -XDELETE https://{IP address}:{port}/ipxapi/server/v1/users/{ID}/smdr -v -k -H "Authorization: bearer $variable" --data-urlencode "{ \"\$and\":[{\"Index\":1},{\"Index\":3},{\"Index\":5}] }"

7) Delete index from 2 to 5

curl -XDELETE https://{IP address}:{port}/ipxapi/server/v1/users/smdr/smdr -v -k -H "Authorization: bearer $variable" --data-urlencode "{\"\$>=\":{\"Index\":2},\"\$<\":{\"Index\":6}}"

5.1.2. Service Example

1) Request data periodically and delete it

If you don't need real time data receiving, you can use this way. You only need to periodically get and delete the requested and received data.

2) Receive events and delete it

If you want to receive data in real time, you can receive via WebSocket session and delete it by delete request. If the data is full because it is not deleted, an alarm may occur depending on the system configuration. It may also be missing because the data is no longer stored.

.

5.2. ACD

“ACD Manager Print” must be enabled in System Data->System Attributes(160~161). API License – AMP license is required. The account for the server can receive ACD data in real time when the WebSocket is connected.

Feature	ACD
Description	ACD event
Service Type	Server
Method	Event
URI	wss://{IP ADDR}:{PORT}/ipxapi/server/v1/users/{User ID}
Header element	Authorization : bearer {token from system} Upgrade: WebSocket
Request Data	

Event Data example	{
	"event":"acd",
	"data":{
	"time":"2020/01/07 14:47:04",
	“callRefID":2,
	"eventCode":18,
	"eventString":"STATION IDLE",
	"sourceType":"Station",
	"sourceNumber":"1004",
	"destType":"",
	"destNumber":" ",
	"data1":" ",
	"data2":" "
	}
	}

Event Data Description	time: YYYY/MM/DD hh:mm:ss, event occurred time callRefID: call reference ID eventCode: see below table eventString: see below table sourceType: CO or Station sourceNumber: source CO or Station Number destType: CO or Station destNumber: destination CO or Station Number data1,data2: See below table

<table>
<tr><th>Event Code</th><th>Event String</th><th>Description</th></tr>
<tr><td>01</td><td>CO BUSY</td><td>CallRefID = Number
Event Code = 1
Source = CO
Destination = STATION/CO
Data1 = N/A
Data2 = N/A</td></tr>
<tr><td>02</td><td>CO IDLE</td><td>CallRefID = N/A
Event Code = 2
Source = CO
Destination = N/A
Data1 = N/A
Data2 = N/A</td></tr>
<tr><td>03</td><td>CO IN RING</td><td>CallRefID = Number
Event Code = 3
Source = CO
Destination = STATION (When group is ringing)
Data1 = Group Bin Number or blank (When group is ringing)
Data2 = Type of Ringing
<table>
<tr><th>Ringing Type</th><th>Value</th></tr>
<tr><td>Ring Speed Dial</td><td>0</td></tr>
<tr><td>Ring One or More Phones</td><td>2</td></tr>
<tr><td>Ring Hunt Group</td><td>4</td></tr>
<tr><td>Ring Voice mail Group</td><td>3</td></tr>
<tr><td>Ring UCD Group</td><td>5</td></tr>
<tr><td>Ring To Modem</td><td>6</td></tr>
</table></td></tr>
<tr><td>11</td><td>AGENT PRIMARY LOGIN</td><td>CallRefID = N/A
Event Code = 11
Source = STATION
Destination = N/A
Data1 = GROUP Bin Number
Data2 = User ID (0000 – 9999)</td></tr>
<tr><td>12</td><td>AGENT PRIMARY LOGOUT</td><td>CallRefID = N/A
Event Code = 12
Source = STATION
Destination = N/A</td></tr>
</table>

		Data1 = GROUP Bin Number Data2 = N/A
13	AGENT UCD UNAVAIL	CallRefID = N/A Event Code = 13 Source = STATION Destination = N/A Data1 = Blank or Dialed Digit: 1 ~ 9, 0(10), *(11), #(12) Data2 = GROUP Bin Number
14	AGENT UCD AVAIL	CallRefID = N/A Event Code = 14 Source = STATION Destination = N/A Data1 = Blank or Dialed Digit: 1 ~ 9, 0(10), *(11), #(12) Data2 = GROUP Bin Number
15	AGENT UCD WRAP UP START	CallRefID = N/A Event Code = 15 Source = STATION Destination = N/A Data1 = N/A Data2 = N/A
16	AGENT UCD WRAP UP END	CallRefID = N/A Event Code = 16 Source = STATION Destination = N/A Data1 = N/A Data2 = N/A
17	STATION BUSY	CallRefID = Number or blank Event Code = 17 Source = STATION Destination = STATION/CO or Blank Data1 = Blank or Dialed Digit: 1 ~ 9, 0(10), *(11), #(12) Data2 = N/A
18	STATION IDLE	CallRefID = N/A Event Code = 18 Source = STATION Destination = N/A Data1 = N/A Data2 = N/A

<table>
<tr><td>19</td><td>STATION DND ACTIVE</td><td>CallRefID = N/A
Event Code = 19
Source = STATION
Destination = N/A
Data1 = N/A
Data2 = N/A</td></tr>
<tr><td>20</td><td>STATION DND INACTIVE</td><td>CallRefID = N/A
Event Code = 20
Source = STATION
Destination = N/A
Data1 = N/A
Data2 = N/A</td></tr>
<tr><td>21</td><td>STATION FORWARD ACTIVE</td><td>CallRefID = N/A
Event Code = 21
Source = STATION
Destination = N/A
Data1 = Forward Type
Data2 = N/A
<table><tr><td>Station Call Forward Type</td><td>Value</td></tr><tr><td>Station Unconditional</td><td>1</td></tr><tr><td>Station No Answer</td><td>17</td></tr><tr><td>Station Busy</td><td>33</td></tr><tr><td>Station No Answer or Busy</td><td>49</td></tr><tr><td>Station Offnet</td><td>2</td></tr><tr><td>CO Line Offnet</td><td>3</td></tr><tr><td>CO Group Offent</td><td>4</td></tr><tr><td>CO All Offent</td><td>5</td></tr><tr><td>UCD Unconditional</td><td>6</td></tr><tr><td>UCD No Answer</td><td>22</td></tr><tr><td>UCD Busy</td><td>38</td></tr><tr><td>UCD No Answer or Busy</td><td>54</td></tr><tr><td>Hunt Unconditional</td><td>7</td></tr><tr><td>Hunt No Answer</td><td>23</td></tr><tr><td>Hunt Busy</td><td>39</td></tr><tr><td>Hunt No Answer or Busy</td><td>55</td></tr><tr><td>Voice Mail Unconditional</td><td>8</td></tr><tr><td>Voice Mail No Answer</td><td>24</td></tr><tr><td>Voice Mail Busy</td><td>40</td></tr><tr><td>Voice Mail No Answer or Busy</td><td>56</td></tr></table></td></tr>
</table>

22	STATION FORWARD INACTIVE	CallRefID = N/A Event Code = 22 Source = STATION Destination = N/A Data1 = N/A Data2 = N/A
23	UCD SUPERVISOR LOGIN	CallRefID = N/A Event Code = 23 Source = STATION Destination = N/A Data1 = GROUP Bin Number Data2 = User ID (0000 – 9999)
24	UCD SUOERVISOR LOGOUT	CallRefID = N/A Event Code = 24 Source = STATION Destination = N/A Data1 = GROUP Bin Number Data2 = N/A
25	AGENT SECONDARY LOGIN	CallRefID = N/A Event Code = 25 Source = STATION Destination = N/A Data1 = GROUP Bin Number Data2 = User ID (0000 – 9999)
26	AGENT SECONDARY LOGOUT	CallRefID = N/A Event Code = 26 Source = STATION Destination = N/A Data1 = GROUP Bin Number Data2 = N/A
41	SEIZE STATION/CO	CallRefID = Number Event Code = 41 Source = STATION/CO Destination = STATION/CO Data1 = N/A Data2 = N/A
42	SEIZE ACK/CONNECT/ANSWER	CallRefID = Number Event Code = 42

<table>
<tr><td></td><td></td><td>Source = STATION/CO
Destination = STATION/CO
Data1 = N/A
Data2 = N/A</td></tr>
<tr><td>43</td><td>SEIZE NAK/ABANDON</td><td>CallRefID = N/A
Event Code = 43
Source = STATION/CO
Destination = STATION/CO
Data1 = N/A
Data2 = N/A</td></tr>
<tr><td>44</td><td>DISCONNECT</td><td>CallRefID = Number
Event Code = 44
Source = STATION/CO
Destination = STATION/CO
Data1 = 0 is normal, 3 is the disconnection from conference.
1 (Data2 is Disconnect Cause).
Data2 = N/A or Disconnect Cause(00~FF)</td></tr>
<tr><td>45</td><td>CONFERENCE ADD</td><td>CallRefID = Number
Event Code = 45
Source = STATION
Destination = STATION/CO when adding to conference, Blank otherwise
Data1 = Conference Action
Data2 = N/A
<table><tr><td>Conference Action Type</td><td>Value</td></tr><tr><td>Terminate Conference</td><td>1</td></tr><tr><td>Enter Conference</td><td>2</td></tr><tr><td>Exit Conference</td><td>3</td></tr></table></td></tr>
<tr><td>46</td><td>CONFERENCE PARK</td><td>Not yet implemented.
CallRefID = Number
Event Code = 46
Source = STATION
Destination = STATION/CO
Data1 = N/A
Data2 = N/A</td></tr>
<tr><td>47</td><td>EXECUTIVE OVERRIDE</td><td>Not yet implemented.
CallRefID = Number
Event Code = 47</td></tr>
</table>

		Source = STATION Destination = STATION Data1 = N/A Data2 = N/A
48	FORWARD	CallRefID = Number Event Code = 48 Source = STATION Destination = STATION/CO Data1 = N/A Data2 = N/A
49	HOLD	CallRefID = Number Event Code = 49 Source = STATION/CO Destination = STATION/CO Data1 = N/A Data2 = N/A
50	HOLD RECALL	CallRefID = Number Event Code = 50 Source = STATION Destination = STATION/CO Data1 = Blank or 0 Data2 = Blank or 0
51	HUNT GROUP	CallRefID = Number Event Code = 51 Source = STATION Destination = STATION Data1 = GROUP Bin Number Data2 = STATION Order Number
52	PAGE ALL	CallRefID = N/A Event Code = 52 Source = STATION Destination = N/A Data1 = N/A Data2 = N/A
53	PAGE EXT	CallRefID = N/A Event Code = 53 Source = STATION Destination = N/A

		Data1 = ZONE Offset Data2 = N/A
54	PAGE INT	CallRefID = N/A Event Code = 54 Source = STATION Destination = N/A Data1 = ZONE Offset Data2 = N/A
55	PARK GROUP	CallRefID = Number Event Code = 55 Source = STATION Destination = STATION/CO Data1 = GROUP Bin Number Data2 = N/A
56	PARK PERSONAL	Not yet implemented. CallRefID = Number Event Code = 56 Source = STATION Destination = STATION/CO Data1 = N/A Data2 = N/A
57	PICK UP	CallRefID = N/A Event Code = 57 Source = STATION Destination = STATION/CO Data1 = N/A Data2 = N/A
58	TRANSFER HOLD	CallRefID = Number Event Code = 58 Source = STATION Destination = STATION/CO Data1 = N/A Data2 = N/A
59	UCD GROUP	CallRefID = Number Event Code = 59 Source = STATION Destination = Number Data1 = GROUP Bin Number

		Data2 = STATION Order Number
60	UCD NO ANSWER TO	Not yet implemented. CallRefID = Number Event Code = 60 Source = STATION Destination = CO Data1 = N/A Data2 = N/A
61	UCD OVERFLOW TO	CallRefID = N/A Event Code = 61 Source = STATION/CO Destination = STATION Data1 = GROUP Bin Number Data2 = N/A
62	UCD QUE HOLD TO	CallRefID = Number Event Code = 62 Source = STATION/CO Destination = N/A Data1 = ACD GROUP Bin Number (0 indicate it is not a ACD call) Data2 = N/A
63	VM GROUP	CallRefID = Number Event Code = 63 Source = STATION Destination = N/A Data1 = GROUP Bin Number Data2 = STATION Bin Number
64	UCD HELP	CallRefID = Number Event Code = 64 Source = STATION Destination = STATION Data1 = GROUP Bin Number Data2 = N/A
65	UCD CALL QUALIFIER	Not yet implemented. CallRefID = Number Event Code = 65 Source = STATION Destination = CO Data1 = N/A

		Data2 = Qualification Code(0000-9999)
66	STATION RING	CallRefID = Number Event Code = 66 Source = STATION Destination = STATION/CO Data1 = ACD GROUP Bin Number (0 indicate it is not a ACD call) Data2 = ACD GROUP Number of Destination Station
67	NUMBER DIALED	CallRefID = Number Event Code = 67 Source = STATION Destination = CO Data1 = N/A Data2 = Number Dialed (Maximum of 32 digits qualified 30 sec. After last digit is dialed)
68	TIME & DATE	CallRefID = N/A Event Code = 68 Source = N/A Destination = N/A Data1 = N/A Data2 = MMDDYY
69	SMDR	CallRefID = Number Event Code = 69 Source = STATION Destination = CO Data1 = HHMMSS (Call Duration) Data2 = Number Dialed (Maximum of 32 digits qualified 30 sec. After last digit is dialed)
70	TRANSFER COMPLETE	CallRefID = Number Event Code = 70 Source = STATION/CO Destination = STATION/CO Data1 = N/A Data2 = N/A
71	INCOMMING DID INFO	CallRefID = Number Event Code = 71 Source = CO Destination = STATION Data1 = DID Number (MAX 7 characters)

		Data2 = N/A
72	INCOMMING CALLER INFO	CallRefID = Number Event Code = 72 Source = CO Destination = STATION Data1 = Phone Number (MAX 18 characters) Data2 = Name (Max 24 characters)
73	CONNECT TX VOICE INFO	CallRefID = Number Event Code = 73 Source = All Destination = All Data1 = High 2 byte of SSRC (4 hex characters) Data2 = low 2 byte of SSRC (4 hex characters)
74	CONNECT RX VOICE INFO	CallRefID = Number Event Code = 74 Source = All Destination = All Data1 = High 2 byte of SSRC (4 hex characters) Data2 = low 2 byte of SSRC (4 hex characters)
75	CONNECT TX VIDEO INFO	CallRefID = Number Event Code = 75 Source = All Destination = All Data1 = High 2 byte of SSRC (4 hex characters) Data2 = low 2 byte of SSRC (4 hex characters)
76	CONNECT RX VIDEO INFO	CallRefID = Number Event Code = 76 Source = All Destination = All Data1 = High 2 byte of SSRC (4 hex characters) Data2 = low 2 byte of SSRC (4 hex characters)
77	ACD INFO CHANGED	CallRefID = N/A Event Code = 77 Source = STATION(System Attendant No) Destination = N/A Data1 = GROUP Bin Number Data2 = N/A
80	INCOMMING DIGIT INFO	CallRefID = Number

		Event Code = 80 Source = CO Destination = Data1 = DIGIT Number (MAX 1 characters) Data2 =
81	AGENT HOTDESK LOGIN	CallRefID = N/A Event Code = 81 Source = STATION (Hotdesk Agent Number that is logged in) Dest = STATION(Dummy Station Number that is logged out) Data1 = N/A Data2 = N/A This Message is sent to application, when a hot desk station is logged in.
82	AGENT OUT OF SERVICE	CallRefID = N/A Event Code = 82 Source = STATION (Out Of Service for all station) Destination = N/A Data1 = N/A Data2 = N/A This Message is sent to application, when a station is turn to Out Of Service status
83	AGENT HOTDESK LOGOUT	CallRefID = N/A Event Code = 83 Source = STATION (Hotdesk Agent Number that is logged out) Dest = STATION (Dummy Station Number that is logged in) Data1 = N/A Data2 = N/A This Message is sent to application, when a hot desk station is logged out.
84	AGENT IN SERVICE	CallRefID = N/A Event Code = 84 Source = STATION (In Service for all station) Destination = N/A Data1 = N/A Data2 = N/A This Message is sent to application, when a station is turn to in service status
85	SMDR ACCOUNT CODE	CallRefID = Number

		Event Code = 85 Source = N/A Destination = N/A Data1 = Account Code(Max 12 digits) Data2 = When SMDR(69) message is sent, if Account Code is available, this message is also sent.
86	SMDR AUTHORIZATION CODE	CallRefID = Number Event Code = 86 Source = N/A Destination = N/A Data1 = Authorization Code(Max 12 digits) Data2 = When SMDR(69) message is sent, if Authorization Code is available, this message is also sent.
87	SMDR COST	(define only, use later) CallRefID = Number Event Code = 87 Source = N/A Destination = N/A Data1 = Cost(Max 10 digits) Data2 =
88	SMDR INCOMING	(define only, use later) CallRefID = Number Event Code = 88 Source = CO Destination = Station Data1 = HHMMSS (Call Duration or Ring Duration) Call Duration or Ring Duration is decided by Call Type. Data2 = Caller Number (Maximum of 23 digits)
89	SMDR ICM	(define only, use later) CallRefID = Number Event Code = 89 Source = Destination = Calling Station Data1 = HHMMSS (Call Duration) Data2 = Called Station

90	SMDR CALL TYPE	(define only, use later) CallRefID = Number Event Code = 90 Source = CO Destination = Station Data1 = Call Type (0000 ~ 0015) Data2 = Call Type is defined as below. 0 : SMDR Off hook 1 : SMDR Incoming Call Transfer 2 : SMDR Incoming Call 3 : SMDR Outgoing Call 4 : SMDR Outgoing Call Transfer 5 : SMDR Hold Call Lost 6 : SMDR Ringing Call Lost 7 : SMDR Group Call Lost 8 : SMDR ICM Call 9 : SMDR ICM SMS 10 : SMDR ICM Paging 11 : SMDR Outgoing Unanswered 12 : SMDR Bar Charge 13 : SMDR Voice Mail 14 : SMDR Net Outgoing 15 : SMDR Net Incoming

Table 3 ACD Event Specification

5.3. CTI (Call Control)

API License – CTI license is required. The account for the server can receive CTI event data in real time when the WebSocket is connected.

API License – CTI-EXT license is required for controlling extensions via this API. And **"Call Control Web API Usage"** must be enabled to control the terminal in Station Data->Common Attributes(111).
If you want to use callRefID, "**CALL TRACKING Enable**" in System Data->System Attributes(160~161) must be enabled.

5.3.1. Term Description

- Web Client: client connecting to web API service
- callRefID: This is the call identifier number assigned when a new call is generated in the iPECS system. It is used to distinguish the call that is activated by Call Event or Call Command. The data type is a positive integer.
- deviceNumber: Number had a Hardware Phone Device.
- serviceDN: Master number of personal groups
- calledDN, callingDN: the number of the called party and the caller.
- agentId – agentId used to log in to the ACD Group be set in advance in iPECS system.
- transactionID: If the client sends a request with transactionID, it will return it in response. Can be used when needed. The data type is a positive integer.

5.3.2.Call Event Start and Stop Request

Web Client can send Monitor Start to request the transmission of the status information of the Extension and send the Monitor Stop to stop sending the status information of the Extension. Web Client receives the status of the Extension

The CTI Command for status message transmission control of the terminal is as follows.

- Monitor Start
- Monitor Stop
- Monitor Status Get

5.3.2.1. Monitor Start

Function Description

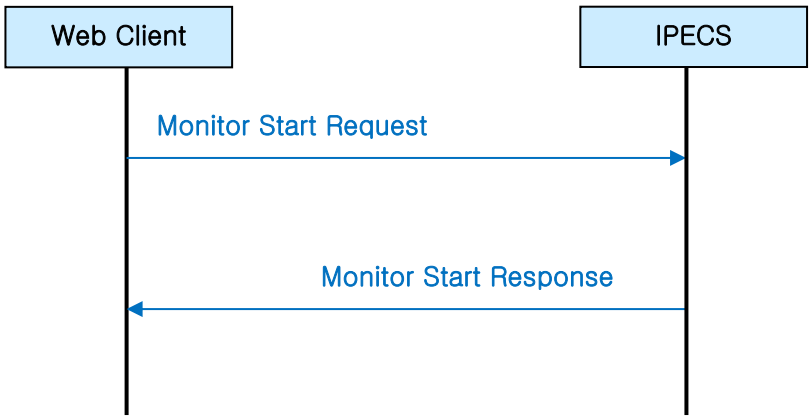
Web Client requests CTI event generation to Extension for status report.

License

API License – CTI required

Usage

Web Client sends a Monitor Start Service Request to the IPECS SYSTEM. The IPECS SYSTEM issues a positive response if the requesting Command is valid, and a Negative Response if it is not.



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/monitor
Method	**POST**
Header	Authorization: Bearer <token from login response>
Parameter	"deviceNumber": [“1000”,”1001”,…,”9999”] }

Positive Response	HTTP 200 OK
	Content:
	{
	“result”:[
	{“1000”: ”OK”},
	{“1001”: “generic”},
	…
	{“9999”: “invalidParameterValue"}
	]
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

The result code is explained in Table 5. Error Code Table.

[Condition]

- The deviceNumber count of Monitor Start can’t be over 100.
- To request more devices, other Monitor Start request should be sent after receiving response of the request.
- The deviceNumber list can be obtained with a Get Terminal Information request.

5.3.2.2. Monitor Stop

Function Description

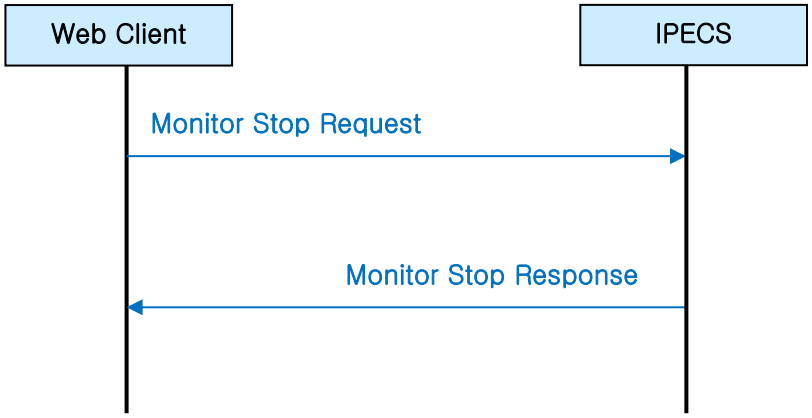
Web Client requests to stop CTI event generation of Extension status.

License

API License – CTI required

Usage

Web Client sends a Monitor Stop Service Request to the IPECS system. The IPECS system issues a positive response if the requesting Command is valid, and a negative response if it is not.



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/monitor
Method	**DELETE**
Header	Authorization: Bearer <token from login response>

Parameter	{
	"deviceNumber": [“1000”,”1001”,…,”9999”]
	}

Positive Response	HTTP 200 OK
	Content:
	{
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	“result”:[
	{“1000”: ”OK”},
	{“1001”: “generic”},
	…
	{“9999”: “invalidParameterValue"}
	]
	}

The result code is explained in Table 5. Error Code Table

5.3.2.3. Get Monitor Status

Function Description

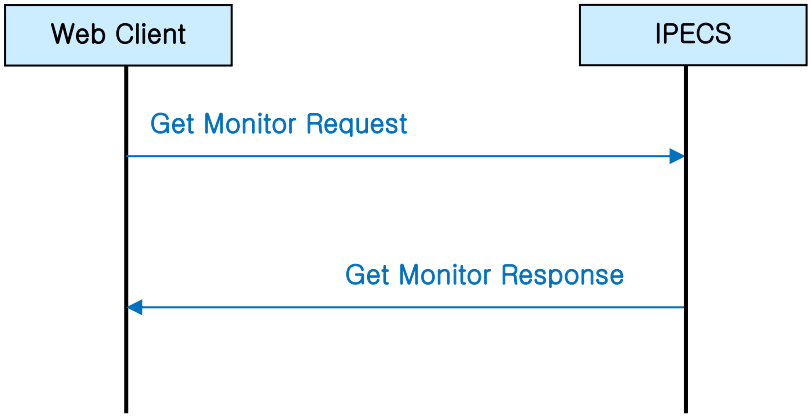
Web Client request to get monitor status information.

License

API License – CTI required

Usage

Web Client sends Get Monitor Status to iPECS system, and iPECS system issues Positive Response if the request command is valid, and Negative Response if it is not.



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/monitor
Method	GET
Header	Authorization: Bearer <token from login response>
Parameter	• startIndex • count

Positive Response	HTTP 200 OK
	Content:
	{
	"data":{
	"monitors":[
	{
	"deviceIndex": 1,
	"deviceNumber": "1000",
	"state": "enable"
	},
	…]
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

Request Parameter

Parameter	Value	Mandatory/ Optional	Description
startIndex	Positive integer	M	Terminal physical number
count	Positive integer	M	Terminal number for request, maximum 100

Response Parameter

Parameter	Value	Mandatory/ Optional	Description

deviceIndex	Positive integer	M	Terminal physical number
deviceNumber	Digit string	M	Terminal number
state	String	M	Monitor state, enable/disable

5.3.2.4. Call Event Message

Function Description

iPECS System sends following CTI events for the Monitor Start requested extensions.

Cause code	Text
3	user_busy
4	call_back
6	forward_immediate
7	forward_busy
8	forward_no_answer
9	forward
10	no_answer
11	pickup
13	invalid_destination
14	dnd
22	new_call
23	no_available_agents
27	recall
28	redirected
31	silent_monitoring
32	screened_transfer
33	trunk_busy
35	blocked
37	consultation
48	normal_clearing
52	blind_transfer
56	timeout
57	conference
58	reroute
59	trunk_call
60	unscreened_transfer
61	bridge_answer
62	branch_answer
63	branch_release
65	group_call
99	callwait
101	mobile_answer
102	madn_answer
103	reject

Table 4 Cause code of Call Event

5.3.2.4.1. Dialing Event

Event to notify the dialing digit of the caller after off-hook

Parameter	Value	Mandatory/ Optional	Description
type	String	M	dialing
deviceNumber	Digit string	M	Terminal number
serviceDN	Digit string	M	Master number of Personal Group(Terminal number by default)
callRefID	Positive integer	M	call reference identifier
evtSerialNum	Positive integer	M	Event serial number

Event:

```
{
   "event": "call",
   "data": {
      "type": "dialing",
      "deviceNumber": "601206",
      "serviceDN": "601206",
      "callRefID": 10,
      "evtSerialNum": 42
   }
}
```

5.3.2.4.2. Proceeding Event

Event to notify the call proceeding status of the caller after completion of dialing

Parameter	Value	Mandatory/ Optional	Description
type	String	M	proceeding
deviceNumber	Digit string	M	Terminal number
serviceDN	Digit string	M	Master number of Personal Group(Terminal number by default)
dialDigit	Digit string	M	Dialed number
callRefID	Positive integer	M	Call reference identifier
evtSerialNum	Positive integer	M	Event serial number

Event:

```
{
   "event": "call",
   "data": {
      "type": "proceeding",
      "deviceNumber": "601206",
      "serviceDN": "601206",
      "dialDigit": "903180551204#",
      "callRefID": 14,
      "evtSerialNum": 57
   }
}
```

5.3.2.4.3. Alerting Event

Event to notify the alerting status of the caller

Parameter	Value	Mandatory/ Optional	Description
type	string	M	alerting
deviceNumber	Digit string	M	Terminal number
serviceDN	Digit string	M	Master number of Personal Group(Terminal number by default)
callingDN	Digit string	M	Calling DN
calledDN	Digit string	M	Called DN
causeCode	Int	M	Refer to <Table 4>
causeMsg	string	M	Refer to <Table 4>
callRefID	Positive integer	M	Call reference identifier
evtSerialNum	Positive integer	M	Event serial number

Event:

```
{
   "event": "call",
   "data": {
      "type": "alerting",
      "deviceNumber": "601206",
      "serviceDN": "605006",
      "callingDN": "605006",
      "calledDN": "+03180551204",
      "causeCode": 22,
      "causeMsg": "new_call",
      "callRefID": 14,
      "evtSerialNum": 58
   }
}
```

5.3.2.4.4. Ringing Event

Event to notify the ringing status of the called

• In order to inform Web Client that a call has been redirected, callType of the Ringing Event is set to 'transfer'.

Parameter	Value	Mandatory/ Optional	Description
type	string	M	ringing
deviceNumber	Digit string	M	Terminal number
serviceDN	Digit string	M	Master number of Personal Group(Terminal number by default)
callingDN	Digit string	M	Calling DN
calledDN	Digit string	M	Called DN
diversionNum	Digit string	O	Number who redirect the call
causeCode	Int	M	Refer to <Table 4>
causeMsg	string	M	Refer to <Table 4>
callRefID	Positive integer	M	Call reference identifier

evtSerialNum	Positive integer	M	Event serial number

Event:

```
{
   "event": "call",
   "data": {
      "type": "ringing",
      "deviceNumber": "605108",
      "serviceDN": "605001",
      "callingDN": "601206",
      "calledDN": "605001",
      "causeCode": 22,
      "causeMsg": "new_call",
      "callRefID": 17,
      "evtSerialNum": 68
   }
}
```

5.3.2.4.5. Connected Event

Event to notify the connect status

Parameter	Value	Mandatory/ Optional	Description
type	string	M	connected
deviceNumber	Digit string	M	Terminal number
serviceDN	Digit string	M	Seized DN on the phone
callingDN	Digit string	M	Calling DN
calledDN	Digit string	M	Called DN
diversionNum	Digit string	O	Number who redirect the call
causeCode	Int	M	Refer to <Table 4>
causeMsg	string	M	Refer to <Table 4>
callRefID	Positive integer	M	Call reference identifier
evtSerialNum	Positive integer	M	Event serial number

Event:

```
{
   "event": "call",
   "data": {
      "type": “connected",
      "deviceNumber": "601207",
      "serviceDN": "605001",
      "callingDN": "601206",
      "calledDN": "605001",
      “diversionNum”: “”
      "causeCode": 22,
      "causeMsg": "new_call",
      "callRefID": 17,
      "evtSerialNum": 71
   }
}
```

5.3.2.4.6. Released Event

Event to notify call disconnection

In case of following conditions, the cause value of the Released Event is notified to Web Client, so Web Client can manage missed call notification and call history.
- When the Mobile extension responds the incoming call
- When the other member of personal group answers the incoming call
- When the incoming call is rejected by the user

Parameter	Value	Mandatory/ Optional	Description
type	string	M	released
deviceNumber	Digit string	M	Terminal number
serviceDN	Digit string	M	Master number of Personal Group(Terminal number by default)
callRefID	Positive integer	M	Call reference identifier
releasedDN	Digit string	M	Released DN
causeCode	Int	M	Refer to <Table 4>
causeMsg	string	M	Refer to <Table 4>
evtSerialNum	Positive integer	M	Event serial number

Event:

```
{
   "event": "call",
   "data": {
      "type": "released",
      "deviceNumber": "601206",
      "serviceDN": "601206",
      "releasedDN": "601206",
      "causeCode": 48,
      "causeMsg": "normal_clearing",
      "callRefID": 16,
      "evtSerialNum": 65
   }
}
```

5.3.2.4.7. ReleasedAsc Event

An event to notify that the party you are calling ended the call.

Parameter	Value	Mandatory/ Optional	Description
type	string	M	releasedasc
deviceNumber	Digit string	M	Terminal number
serviceDN	Digit string	M	Master number of Personal Group(Terminal number by default)
releasedAscDN	Digit string	M	Released DN
causeCode	Int	M	Refer to <Table 4>
causeMsg	string	M	Refer to <Table 4>
callRefID	Positive integer	M	Call reference identifier
evtSerialNum	Positive integer	M	Event serial number

Event:

```
{
   "event": "call",
   "data": {
      "type": "releasedasc",
      "deviceNumber": "601206",
      "serviceDN": "601206",
      "releasedAscDN": "605001",
      "causeCode": 48,
      "causeMsg": "normal_clearing",
      "callRefID": 17,
      "evtSerialNum": 74
   }
}
```

5.3.2.4.8. Rerouted Event

Event to notify that the call diverted to another number and is no longer available

Parameter	Value	Mandatory/ Optional	Description
type	string	M	rerouted
deviceNumber	Digit string	M	Terminal number
serviceDN	Digit string	M	Master number of Personal Group(Terminal number by default)
calledDN	Digit string	M	Called DN
reroutingDN	Digit string	M	Calling DN
causeCode	Int	M	Refer to <Table 4>
causeMsg	string	M	Refer to <Table 4>
callRefID	Positive integer	M	Call reference identifier
evtSerialNum	Positive integer	M	Event serial number

Event:

```
{
   "event": "call",
   "data": {
      "type": "rerouted",
      "deviceNumber": "601206",
      "serviceDN": "601206",
      "calledDN": "601208",
      "reroutingDN": "601206",
      "causeCode": 58,
      "causeMsg": "reroute",
      "callRefID": 18,
      "evtSerialNum": 83
   }
}
```

5.3.2.4.9. Rerouting Event

Rerouting Event can be sent in ringing and connected state.

Parameter	Value	Mandatory/ Optional	Description
type	string	M	rerouting
deviceNumber	Digit string	M	Terminal number
serviceDN	Digit string	M	Master number of Personal Group(Terminal number by default)
calledDN	Digit string	M	Called DN
reroutingDN	Digit string	M	Calling DN
causeCode	Int	M	Refer to <Table 4>
causeMsg	string	M	Refer to <Table 4>
callRefID	Positive integer	M	Call reference identifier
evtSerialNum	Positive integer	M	Event serial number

Event:

```
{
  "event": "call",
  "data": {
    "type": "rerouting",
    "deviceNumber": "601206",
    "serviceDN": "601206",
    "calledDN": "601208",
    "reroutingDN": "601206",
    "causeCode": 58,
    "causeMsg": "reroute",
    "callRefID": 18,
    "evtSerialNum": 83
  }
}
```

5.3.2.4.10. Failed Event

Event to notify that the call is failed

Parameter	Value	Mandatory/ Optional	Description
type	string	M	fail
deviceNumber	Digit string	M	Terminal number
serviceDN	Digit string	M	Master number of Personal Group(Terminal number by default)
callingDN	Digit string	M	Calling DN
calledDN	Digit string	M	Called DN
busySvcList	string	O	Busy service list callback, campon, callwait, intrude Ex)“busySvcList”:[“callback”, “campon”]
causeCode	Int	M	Refer to <Table 4>
causeMsg	string	M	Refer to <Table 4>
callRefID	Positive integer	M	Call reference identifier
evtSerialNum	Positive integer	M	Event serial number

Event:

```
{
```

```
  "event": "call",
  "data": {
    "type": "fail",
    "deviceNumber": "601207",
    "serviceDN": "601207",
    "callingDN": "601207",
    "calledDN": "601208",
    "busySvcList": ["intrusion", "callback", "callwait"],
    "causeCode": 3,
    "causeMsg": "user_busy",
    "callRefID": 26,
    "evtSerialNum": 143
  }
}
```


5.3.2.4.11. Held Event

Event to notify that call is held

Parameter	Value	Mandatory/ Optional	Description
type	string	M	held
deviceNumber	Digit string	M	Terminal number
serviceDN	Digit string	M	Master number of Personal Group(Terminal number by default)
holdDN	Digit string	M	Holding DN
callRefID	Positive integer	M	Call reference identifier
evtSerialNum	Positive integer	M	Event serial number

Event:


```
{
  "event": "call",
  "data": {
    "type": " held",
    "deviceNumber": "601206",
    "serviceDN": "601206",
    "holdDN": "601207",
    "callRefID": 29,
    "evtSerialNum": 170
  }
}
```


5.3.2.4.12. Hold Event

Event to notify that the user hold the call

Parameter	Value	Mandatory/ Optional	Description
type	string	M	hold
deviceNumber	Digit string	M	Terminal number
serviceDN	Digit string	M	Master number of Personal Group(Terminal number by default)
holdDN	Digit string	M	Holding DN

callRefID	Positive integer	M	Call reference identifier
evtSerialNum	Positive integer	M	Event serial number

Event:

```
{
   "event": "call",
   "data": {
      "type": " hold",
      "deviceNumber": "601207",
      "serviceDN": "601207",
      "holdDN": "601207",
      "callRefID": 29,
      "evtSerialNum": 169
   }
}
```

5.3.2.4.13. Retrieved Event

Event to notify that the held call is retrieved

• Retrieved Event is sent to retrieved device

Parameter	Value	Mandatory/ Optional	Description
type	string	M	retrieved
deviceNumber	Digit string	M	Terminal number
serviceDN	Digit string	M	Master number of Personal Group(Terminal number by default)
retrieveDN	Digit string	M	Retrieve DN
callRefID	Positive integer	M	Call reference identifier
evtSerialNum	Positive integer	M	Event serial number

Event:

```
{
   "event": "call",
   "data": {
      "type": "retrieved",
      "deviceNumber": "601206",
      "serviceDN": "601206",
      "retrieveDN": "601207",
      "callRefID": 29,
      "evtSerialNum": 172
   }
}
```

5.3.2.4.14. Retrieve Event

Event to notify that the user retrieved the held call

• Retrieve Event is sent to retrieving device

Parameter	Value	Mandatory/ Optional	Description

type	string	M	retrieve
deviceNumber	Digit string	M	Terminal number
serviceDN	Digit string	M	Master number of Personal Group(Terminal number by default)
retrieveDN	Digit string	M	Retrieve DN
callRefID	Positive integer	M	Call reference identifier
evtSerialNum	Positive integer	M	Event serial number

Event:

```
{
   "event": "call",
   "data": {
      "type": "retrieve",
      "deviceNumber": "601207",
      "serviceDN": "601207",
      "retrieveDN": "601207",
      "callRefID": 29,
      "evtSerialNum": 171
   }
}
```

5.3.2.4.15. Transferred Event

Event to notify that the call is transferred

• Transferred Event is sent to transferee and transfer-to device

Parameter	Value	Mandatory/ Optional	Description
type	string	M	transferred
deviceNumber	Digit string	M	Terminal number
serviceDN	Digit string	M	Master number of Personal Group(Terminal number by default)
transferToDN	Digit string	M	Transfer target DN
transferorDN	Digit string	M	Transferor DN
callRefID	Positive integer	M	Call reference identifier
evtSerialNum	Positive integer	M	Event serial number

Event:

```
{
   "event": "call",
   "data": {
      "type": "transferred",
      "deviceNumber": "601206",
      "serviceDN": "605006",
      "transferToDN": "605008",
      "transferorDN": "605001",
      "callRefID": 31,
      "evtSerialNum": 197
   }
}
```

5.3.2.4.16. Transferring Event

Event to notify that the call is transferring

• Transferring Event is sent to transferrer device

Parameter	Value	Mandatory/ Optional	Description
type	string	M	transferring
deviceNumber	Digit string	M	Terminal number
serviceDN	Digit string	M	Master number of Personal Group(Terminal number by default)
transferedDN	Digit string	M	Transferred DN
transferToDN	Digit string	M	Transfer target DN
transferorDN	Digit string	M	Transferor DN
callType	string	M	blind unscreened screened
callRefID	Positive integer	M	Call reference identifier
evtSerialNum	Positive integer	M	Event serial number

Event:

```
{
   "event": "call",
   "data": {
      "type": "transferring",
      "deviceNumber": "601207",
      "serviceDN": "605001",
      "transferedDN": "605006",
      "transferToDN": "605008",
      "transferorDN": "605001",
      "callType": "screened",
      "callRefID": 31,
      "evtSerialNum": 194
   }
}
```

5.3.2.4.17. Conferenced Event

Event to notify that conference call is established

• Conferenced Event is sent to all joined member except conference call organizer

Parameter	Value	Mandatory/ Optional	Description
type	string	M	conferenced
deviceNumber	Digit string	M	Terminal number
serviceDN	Digit string	M	Master number of Personal Group(Terminal number by default)
memberDN[…]	list	M	Conference member (max: eMG 13, UCP 32)
masterDN	Digit string	M	Conference call organizer

newConfMemberDN	Digit string	M	New conference member DN
callRefID	Positive integer	M	Call reference identifier
evtSerialNum	Positive integer	M	Event serial number

Event:

```
{
   "event": "call",
   "data": {
      "type": "conferenced",
      "deviceNumber": "601206",
      "serviceDN": "601206",
      "memberDN": ["601206", "601207", "601208"],
      “masterDN”: “601207”,
      "newConfMemberDN": "601208",
      "callRefID": 34,
      "evtSerialNum": 219
   }
}
```

5.3.2.4.18. Conferencing Event

Event to notify that conference call is established (for organizer)

Parameter	Value	Mandatory/ Optional	Description
type	string	M	conferencing
deviceNumber	Digit string	M	Terminal number
serviceDN	Digit string	M	Master number of Personal Group(Terminal number by default)
memberDN[…]	list	M	Conference member (max: eMG 13, UCP 32)
newConfMemberDN	Digit string	M	New conference member DN
callRefID	Positive integer	M	Call reference identifier
evtSerialNum	Positive integer	M	Event serial number

Event:

```
{
   "event": "call",
   "data": {
      "type": "conferencing",
      "deviceNumber": "601207",
      "serviceDN": "601207",
      "memberDN": ["601206", "601207", "601208"],
      "newConfMemberDN": "601208",
      "callRefID": 34,
      "evtSerialNum": 217
   }
}
```

5.3.2.4.19. DoNotDisturb Event

Event to notify that DND status is changed

Parameter	Value	Mandatory/ Optional	Description
type	string	M	dnd
deviceNumber	Digit string	M	Terminal number
serviceDN	Digit string	M	Master number of Personal Group(Terminal number by default)
svcType	string	M	enable, disable
evtSerialNum	Positive integer	M	Event serial number

Event:

```
{
   "event": "call",
   "data": {
      "type": "dnd",
      "deviceNumber": "601207",
      "serviceDN": "601207",
      "svcType": "enable",
      "evtSerialNum": 228
   }
}
```

5.3.2.4.20. Forwarding Event

Event to notify that call forwarding status is changed

Parameter	Value	Mandatory/ Optional	Description
type	string	M	forwarding
deviceNumber	Digit string	M	Terminal number
serviceDN	Digit string	M	Master number of Personal Group(Terminal number by default)
destination	Digit string	M	Call forward target
svcType	string	M	Call forward type disable uncondition busy no_answer busy_noans DND
evtSerialNum	Positive integer	M	Event serial number

Event:

```
{
   "event": "call",
   "data": {
      "type": "forwarding",
      "deviceNumber": "601207",
      "serviceDN": "601207",
      "destination": "1208",
      "svcType": "no_answer",
      "evtSerialNum": 230
   }
}
```

5.3.2.4.21. Registered Event

Event to notify that the device is registered

- This event is sent only for the devices requested Monitor Start.

Parameter	Value	Mandatory/ Optional	Description
type	string	M	registered
deviceNumber	Digit string	M	Terminal number
evtSerialNum	Positive integer	M	Event serial number

Event:

```
{
   "event": "call",
   "data": {
      "type": "registered",
      "deviceNumber": "601207",
      "evtSerialNum": 242
   }
}
```

5.3.2.4.22. UnRegistered Event

Event to notify the unregistered state of the phone to Web Client.

- This event is issued only to the terminal that is requested Monitor Start.
- For terminals that are not connected at Monitor Start, the Monitor Start Response is issued as OK and issues UnRegistered Event. By doing this, you can issue a Registered Event when the terminal is connected, and do not have to restart Monitor Start on Web Client.

Parameter	Value	Mandatory/ Optional	Description
type	string	M	unregistered
deviceNumber	Digit string	M	Terminal number
evtSerialNum	Positive integer	M	Event serial number

Event:

```
{
   "event": "call",
   "data": {
      "type": " unregistered",
      "deviceNumber": "601207",
      "evtSerialNum": 241
   }
}
```

5.3.2.4.23. MWI Event

Event to notify the MWI(Message Wait Indication) information of the phone to Web Client.

Parameter	Value	Mandatory/	Description

		Optional	
type	string	M	mwi
deviceNumber	Digit string	M	Terminal number
serviceDN	Digit string	M	Master number of Personal Group(Terminal number by default)
msgStatus	true, false	M	true : new message exists false : new message doesn't exist
newMsg	Positive integer	M	New message count
oldMsg	Positive integer	M	Old message count
newUrgentMsg	Positive integer	M	New urgent message count
evtSerialNum	Positive integer	M	Event serial number

Event:

```
{
   "event": "call",
   "data": {
      "type": "mwi",
      "deviceNumber": "601207",
      "serviceDN":"601207"
      "msgStatus":"true",
      "newMsg":34
      "oldMsg":2
      "newUrgentMsg":4
      "evtSerialNum": 242
   }
}
```

5.3.2.4.24. ACD Agent LogOn Event

Event to notify agent log on state to Web Client.

Parameter	Value	Mandatory/ Optional	Description
type	string	M	agent_logon
deviceNumber	Digit string	M	Terminal number
serviceDN	Digit string	M	Master number of Personal Group(Terminal number by default)
primaryAgentId	Digit string	M	Unique id on primary ACD group
primaryAcdGroup	Digit string	M	Primary ACD group which agent belongs
secondaryAgentId	Digit string	M	Unique ID on secondary ACD group
secondaryAcdGroup	Digit string	M	Secondary ACD group which agent belongs
evtSerialNum	Positive integer	M	Event serial number

Event:

```
{
   "event": "call",
   "data": {
     "type": "agent_logon",
     "deviceNumber": "601207",
     "serviceDN": "601207",
     "primaryAgentId": "1207"
```

```
    “primaryAcdGroup”: “60304”,
    “secondaryAgentId”: “1207”,
    “secondaryAcdGroup”: “60304”,
    "evtSerialNum": 241
  }
}
```


5.3.2.4.25. ACD Agent LogOff Event

Event to notify agent log off state to Web Client.

Parameter	Value	Mandatory/ Optional	Description
type	string	M	agent_logoff
deviceNumber	Digit string	M	Terminal number
serviceDN	Digit string	M	Master number of Personal Group(Terminal number by default)
primaryAgentId	Digit string	M	Unique id on primary ACD group
primaryAcdGroup	Digit string	M	Primary ACD group which agent belongs
secondaryAgentId	Digit string	M	Unique ID on secondary ACD group
secondaryAcdGroup	Digit string	M	Secondary ACD group which agent belongs
evtSerialNum	Positive integer	M	Event serial number

Event:


```
{
  "event": "call",
  "data": {
    "type": “agent_logoff",
    "deviceNumber": "601207",
    "serviceDN": "601207",
    “primaryAgentId”: “1207”
    “primaryAcdGroup”: “60304”
    “secondaryAgentId”: “1207”
    “secondaryAcdGroup”: “60304”
    "evtSerialNum": 241
  }
}
```


5.3.2.4.26. ACD Agent Ready Event

Event to notify agent Ready state to Web Client.

Parameter	Value	Mandatory/ Optional	Description
type	string	M	agent_ready
deviceNumber	Digit string	M	Terminal number
serviceDN	Digit string	M	Master number of Personal Group(Terminal number by default)
primaryAgentId	Digit string	M	Unique id on primary ACD group
primaryAcdGroup	Digit string	M	Primary ACD group which agent belongs
secondaryAgentId	Digit string	M	Unique ID on secondary ACD group

secondaryAcdGroup	Digit string	M	Secondary ACD group which agent belongs
evtSerialNum	Positive integer	M	Event serial number

Event:

```
{
   "event": "call",
   "data": {
      "type": "agent_ready",
      "deviceNumber": "601207",
      "serviceDN": "601207",
      “primaryAgentId”: “1207”,
      “primaryAcdGroup”: “60304”,
      “secondaryAgentId”: “1207”,
      “secondaryAcdGroup”: “60304”,
      "evtSerialNum": 241
   }
}
```

5.3.2.4.27. ACD Agent Not Ready Event

Event to notify agent Not Ready state to Web Client.

Parameter	Value	Mandatory/ Optional	Description
type	string	M	agent_notready
deviceNumber	Digit string	M	Terminal number
serviceDN	Digit string	M	Master number of Personal Group(Terminal number by default)
primaryAgentId	Digit string	M	Unique id on primary ACD group
primaryAcdGroup	Digit string	M	Primary ACD group which agent belongs
secondaryAgentId	Digit string	M	Unique ID on secondary ACD group
secondaryAcdGroup	Digit string	M	Secondary ACD group which agent belongs
evtSerialNum	Positive integer	M	Event serial number

Event:

```
{
   "event": "call",
   "data": {
      "type": "agent_notready",
      "deviceNumber": "601207",
      "serviceDN": "601207",
      “primaryAgentId”: “1207”
      “primaryAcdGroup”: “60304”,
      “secondaryAgentId”: “1207”,
      “secondaryAcdGroup”: “60304”,
      "evtSerialNum": 241
   }
}
```

5.3.2.4.28. ACD Agent Work Event

Event to notify agent Work after call state to Web Client.

Parameter	Value	Mandatory/ Optional	Description
type	String	M	agent_work
deviceNumber	Digit string	M	Terminal number
serviceDN	Digit string	M	Master number of Personal Group(Terminal number by default)
primaryAgentId	Digit string	M	Unique id on primary ACD group
primaryAcdGroup	Digit string	M	Primary ACD group which agent belongs
secondaryAgentId	Digit string	M	Unique ID on secondary ACD group
secondaryAcdGroup	Digit string	M	Secondary ACD group which agent belongs
evtSerialNum	Positive integer	M	Event serial number

Event:

```
{
   "event": "call",
   "data": {
      "type": "agent_work",
      "deviceNumber": "601207",
      "serviceDN": "601207",
      “primaryAgentId”: “1207”,
      “primaryAcdGroup”: “60304”,
      “secondaryAgentId”: “1207”,
      “secondaryAcdGroup”: “60304”,
      "evtSerialNum": 241
   }
}
```

5.3.2.4.29. ACD Agent Busy Event

Event to notify agent busy state(phone busy)to Web Client.

Parameter	Value	Mandatory/ Optional	Description
type	String	M	agent_busy
deviceNumber	Digit string	M	Terminal number
serviceDN	Digit string	M	Master number of Personal Group(Terminal number by default)
primaryAgentId	Digit string	M	Unique id on primary ACD group
primaryAcdGroup	Digit string	M	Primary ACD group which agent belongs
secondaryAgentId	Digit string	M	Unique ID on secondary ACD group
secondaryAcdGroup	Digit string	M	Secondary ACD group which agent belongs
evtSerialNum	Positive integer	M	Event serial number

Event:

```
{
   "event": "call",
   "data": {
```

```
    "type": "agent_busy",
    "deviceNumber": "601207",
    "serviceDN": "601207",
    “primaryAgentId”: “1207”,
    “primaryAcdGroup”: “60304”,
    “secondaryAgentId”: “1207”,
    “secondaryAcdGroup”: “60304”,
    "evtSerialNum": 241
  }
}
```


5.3.2.4.30. ACD Group Queueing Event

Event to notify ACD Group just got a call and wait route to Web Client.

Parameter	Value	Mandatory/ Optional	Description
type	String	M	queueing
deviceNumber	Digit string	M	ACD Group Number
serviceDN	Digit string	M	Master number of Personal Group(Terminal number by default)
callingDN	Digit string	M	Calling DN
calledDN	Digit string	M	Called DN
diversionNum	Digit string	O	Number who redirect the call
qCount	Positive integer	M	Queued Count on ACD Group
primaryAgentId	Digit string	M	Unique id on primary ACD group
primaryAcdGroup	Digit string	M	Primary ACD group which agent belongs
secondaryAgentId	Digit string	M	Unique ID on secondary ACD group
secondaryAcdGroup	Digit string	M	Secondary ACD group which agent belongs
causeCode	Int	M	Refer to <Table 4>
causeMsg	string	M	Refer to <Table 4>
evtSerialNum	Positive integer	M	Event serial number

Event:


```
{
  "event": "call",
  "data": {
    "type": "queueing",
    "deviceNumber": "60304",
    "serviceDN": "60304",
    "callingDN": “601207”,
    "calledDN": “60304”,
    "diversionNum": “”,
    "qCount": 90,
    “primaryAgentId”: “1207”,
    “primaryAcdGroup”: “60304”,
    “secondaryAgentId”: “1207”,
    “secondaryAcdGroup”: “60304”,
    "causeCode": 23,
    "causeMsg": no_available_agents,
    "evtSerialNum": 241
  }
```

```
}
```

5.3.2.4.31. ACD Group Queued Event

Event to notify caller's call has queued on ACD Group to Web Client.

Parameter	Value	Mandatory/ Optional	Description
type	string	M	queueing
deviceNumber	Digit string	M	Terminal number
serviceDN	Digit string	M	DN
callingDN	Digit string	M	Calling DN
calledDN	Digit string	M	Called DN
qCount	Positive integer	M	Queued Count on ACD Group
primaryAgentId	Digit string	M	Unique id on primary ACD group
primaryAcdGroup	Digit string	M	Primary ACD group which agent belongs
secondaryAgentId	Digit string	M	Unique ID on secondary ACD group
secondaryAcdGroup	Digit string	M	Secondary ACD group which agent belongs
causeCode	Int	M	Refer to <Table 4>
causeMsg	string	M	Refer to <Table 4>
evtSerialNum	Positive integer	M	Event serial number

Event:

```
{
    "type": "queued",
    "deviceNumber": "601207",
    "serviceDN": "601207",
    "callingDN": "601207",
    "calledDN": "60304",
    "qCount": 90,
    "primaryAgentId": "1207",
    "primaryAcdGroup": "60304",
    "secondaryAgentId": "1207",
    "secondaryAcdGroup": "60304",
    "causeCode": 23,
    "causeMsg": no_available_agents,
    "evtSerialNum": 241
}
```

5.3.2.4.32. ConfGroup Member Event

Event to notify Conference room member has added or changed to Web Client

Parameter	Value	Mandatory/ Optional	Description
type	string	M	confgroupmember
groupIndex	Positive integer	M	Conference Group Index
groupUsage	string	M	Conference Group Usage true, false
memberIndex	Positive integer	M	Member Index on Conference Group
memberDN	Digit string	M	Member number on Conference Group

memberState	String	M	Member state, idle or busy
evtSerialNum	Positive integer	M	Event serial number

Event:

```
{
   "event": "call",
   "data": {
      "type": " confgroupmember",
      "groupIndex": 10,
      "groupUsage": “true”,
      "memberIndex": 2,
      "memberDN": “601207”,
      “memberState”: “idle”,
      "evtSerialNum": 241
   }
}
```

5.3.2.4.33. ConfGroup muteOn Event

Information of group call member is Mute or not

Parameter	Value	Mandatory/ Optional	Description
Type	string	M	muteon
deviceNumber	Digit string	M	Terminal Number
serviceDN	Digit string	M	Master number of Personal Group (Terminal number by default)
groupIndex	Positive integer	M	Conference Group Index
memberIndex	Positive integer	M	Member Index on Conference Group
muteOn	string	M	true false
evtSerialNum	Positive integer	M	Event serial number

Event:

```
{
   "event": "call",
   "data": {
      "type": "muteon",
      "deviceNumber": "60700",
      "serviceDN": "60700",
      "groupIndex": 10,
      "memberIndex": 2,
      "muteOn": “false”,
      "evtSerialNum": 241
   }
}
```

5.3.2.4.34. ConfRoom Member Event

Event to notify Conference room member has added or changed to Web Client

Parameter	Value	Mandatory/ Optional	Description
type	string	M	confroommember
deviceNumber	Digit string	M	Conference Room Number
serviceDN	Digit string	M	Conference Room Number
roomIndex	Positive integer	M	Conference Room Index
roomUsage	string	M	Conference Room Usage true, false
memberCount	Positive integer	M	Member count on Conference Room, Max eMG: 13, UCP: 32
member	[Digit string]	M	Member list on Conference Room
evtSerialNum	Positive integer	M	Event serial number

Event:

```
{
   "event": "call",
   "data": {
      "type": "confroommember ",
      "deviceNumber": "603000",
      "serviceDN": "603000",
      "roomIndex": 10,
      "roomUsage": “true”,
      "memberCount": 2,
      "member": [“601207”, “601208”],
      "evtSerialNum": 241
   }
}
```

5.3.2.4.35. ConfRoom Usage Event

Event to notify Conference room has added or changed to Web Client

Parameter	Value	Mandatory/ Optional	Description
type	string	M	confroomusage
deviceNumber	Digit string	M	Conference Room Number
serviceDN	Digit string	M	Conference Room Number
roomIndex	Digit string	M	Conference RoomIndex
roomUsage	Digit string	M	Conference Room Usage true, false
roomPwd	String	M	Conference Room Password
evtSerialNum	Positive integer	M	Event serial number

Event:

```
{
   "event": "call",
   "data": {
      "type": "confroomusage",
      "deviceNumber": "603000",
      "serviceDN": "603000",
```

```
      "roomIndex": 10,
      "roomUsage": “false”,
      "roomPwd": “1234”,
      "evtSerialNum": 241
    }
}
```

5.3.3.Presence Event Start and Stop

This chapter describes messages and procedures that Web Client can request to start or stop the presence information of the DN. When Web Client requests the Presence transmission, the iPECS system start to transmit the status information of DNs. The information will be sent only DNs that the status of terminals changed.

The CTI Command for controlling the status message transmission of the DN is as follows:

- Presence Start
- Presence Stop
- Presence Status Get

5.3.3.1. Presence Start

Function Description

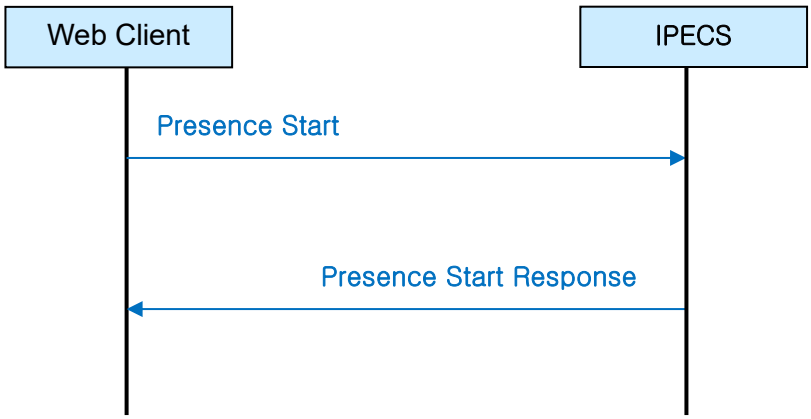
Web Client can send Presence Start Request to receive presence information. Presence Event is transmitted when DNs that the status of terminals changed.

License

API License – CTI required

Usage

Web Client sends a Presence Start Request to the iPECS system, and the iPECS system issues a Positive Response if the request command is valid, and a Negative Response if it is not.



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/presence
Method	POST
Header	Authorization: Bearer <token from login response>
Parameter	{ “deviceNumber”: [“1000”,”1001”, …]

	}

Positive Response	HTTP 200 OK
	Content:
	{
	“result”:[
	{“1000”: ”OK”},
	{“1001”: “generic”},
	…
	{“9999”: “invalidParameterValue"}
	]
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

The result code is explained in Table 5. Error Code Table.

[Condition]

- In case of Positive Response, it issues a Presence Event of the status changed DN.
- The deviceNumber count of Presence Start can’t be over 100.
- To request more devices, other Presence Start request should be sent after receiving response of the request.
- The deviceNumber list can be obtained with a Get Terminal Information request.

In case of Negative Response, return Error value as followings:

- Operation (serviceNotSupported) - service not provided
- Operation (invalidMonitorObjectType) - invalid type
- Operation (invalidDeviceReferenceIdentifier) - invalid deviceNumber

[Limitation]

- Presence information is supported only for extensions.

5.3.3.2. Presence Stop

Function Description

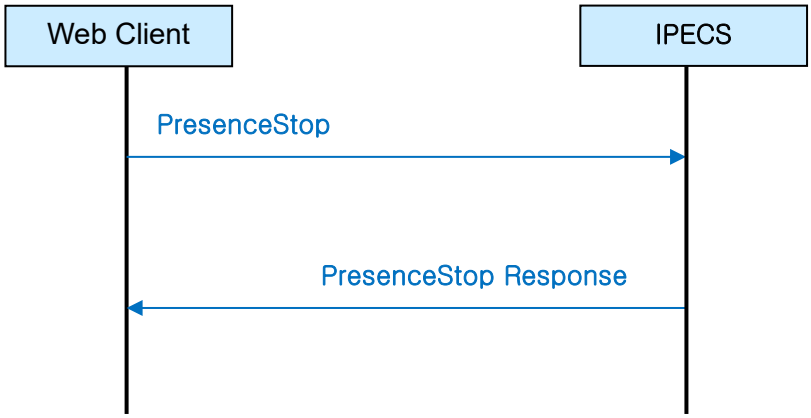
Web Client can send Presence Stop Request to stop presence transmission.

License

API License – CTI required

Usage

Web Client sends a Presence Stop Request to the iPECS system, and the iPECS system issues a Positive Response if the request command is valid, and a Negative Response if it is not.



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/presence
Method	**DELETE**
Header	Authorization: Bearer <token from login response>

Parameter	{
	“deviceNumber”: [“1000”,”1001”, …]
	}

Positive Response	HTTP 200 OK
	Content:
	{
	“result”:[
	{“1000”: ”OK”},
	{“1001”: “generic”},
	…
	{“9999”: “invalidParameterValue"}
	]
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

The result code is explained in Table 5. Error Code Table.

[Condition]

In case of Negative Response, return Error value as below.

- Operation (serviceNotSupported) - service not provided
- Operation (invalidMonitorObjectType) - invalid type
- Operation (invalidDeviceReferenceIdentifier) - invalid deviceNumber

5.3.3.3. Get Presence Status

Function Description

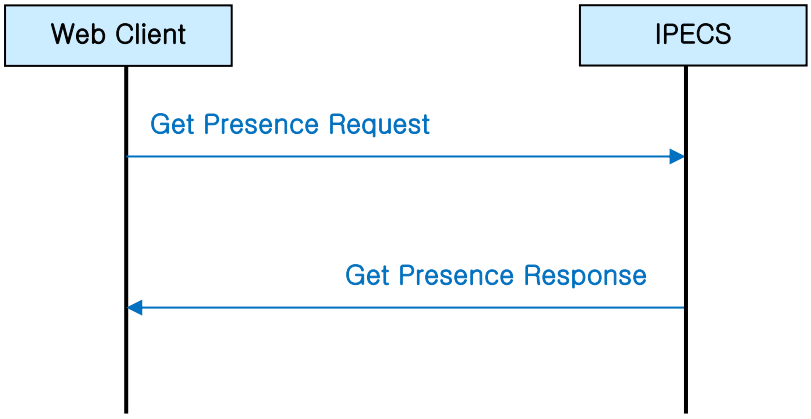
Web Client request to get presence status information.

License

API License – CTI required

Usage

Web Client sends Get Presence Status to iPECS system, and iPECS system issues Positive Response if the request command is valid, and Negative Response if it is not.



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/presence
Method	GET
Header	Authorization: Bearer <token from login response>
Parameter	• startIndex • count

Positive Response	HTTP 200 OK
	Content:
	{
	"data":{
	"presences":[
	{
	"deviceIndex": 1,
	"deviceNumber": "1000",
	"state": "enable"
	},
	…]
	}

Negative Response	HTTP 401 Unauthorized, HTTP 403 Forbidden

	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

Request Parameter

Parameter	Value	Mandatory/ Optional	Description
startIndex	Positive integer	M	Terminal physical number
count	Positive integer	M	Terminal number for request, maximum 100

Response Parameter

Parameter	Value	Mandatory/ Optional	Description
deviceIndex	Positive integer	M	Terminal physical number
deviceNumber	Digit string	M	Terminal number
state	String	M	Presence state, enable/disable

5.3.3.4. Presence Event

Presence Event transmits status information for the DN which status has changed.

Parameter	Value	Mandatory/ Optional	Description
type	string	M	presence
deviceNumber	Digit string	M	Terminal number
deviceType	string	M	ext : extension trk : trunk
status	string	M	block idle DND stn_talk trk_talk busy fwd_all fwd_uncond fwd_busy fwd_nans fwd_bsna ring hold conf
evtSerialNum	Positive integer	M	Event serial number

Event:

```
{
   "event":"call",
   "data":{
      "type":"presence",
      "deviceNumber":"1003",
      "deviceType":"ext",
      "status":"busy",
      "evtSerialNum":1
   }
}
```


[Condition]

The status information of the DN is shown below.

- block: unregistered state
- idle: idle state
- DND: DND setting status
- stn_talk: status of extension call
- trk_talk: Busy state during trunk line
- busy: busy state other than extension / trunk line call
- cfwd_all; Unconditional call forward registration status
- cfwd_busy: call forwarding registration status
- cfwd_nans: No-answer forwarding registration status
- cfwd_bsna: busy / no-answer forwarding registration status
- ring: ringing status
- hold: holding the counter part
- conf: conference status

5.3.4. CO Status Event Start and Stop

This chapter describes messages and procedures that Web Client can request to start or stop the CO status information. When Web Client requests the Check CO transmission, the iPECS system start to transmit the status information of CO. The information will be sent when the status of CO is changed.

The CTI Command for controlling the status message transmission of the DN is as follows:

- Check CO Start
- Check CO Stop
- Check CO Status Get

5.3.4.1. Check CO Start

Function Description

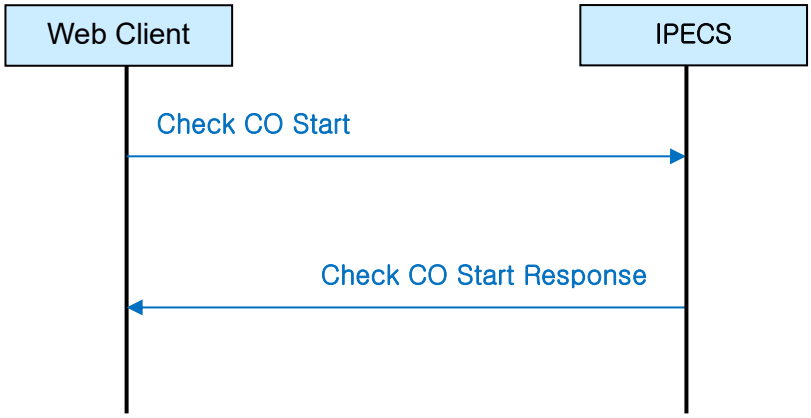
Web Client can send Check CO Start Request to receive CO status information. CO Status Event is transmitted when CO status is changed.

License

API License – CTI required

Usage

Web Client sends a Check CO Start Request to the iPECS system, and the iPECS system issues a Positive Response if the request command is valid, and a Negative Response if it is not.



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/check_co
Method	POST
Header	Authorization: Bearer <token from login response>

Parameter	{
	“deviceNumber”: [“10”,”11”, …]
	}

Positive Response	HTTP 200 OK
	Content:
	{
	“result”:[
	{“10”: ”OK”},
	{“11”: “generic”},
	…
	{“20”: “invalidParameterValue"}
	]
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

The result code is explained in Table 5. Error Code Table.

[Condition]

- In case of Positive Response, it issues a Check CO Event when the CO status is changed.
- The CO count of Check CO Start can't be over 100.
- To request more devices, other Check CO Start request should be sent after receiving response of the request.
- The deviceNumber is CO number.

In case of Negative Response, return Error value as followings:

- Operation (serviceNotSupported) - service not provided
- Operation (invalidMonitorObjectType) - invalid type
- Operation (invalidDeviceReferenceIdentifier) - invalid deviceNumber

[Limitation]

- Presence information is supported only for extensions.

5.3.4.2. Check CO Stop

Function Description

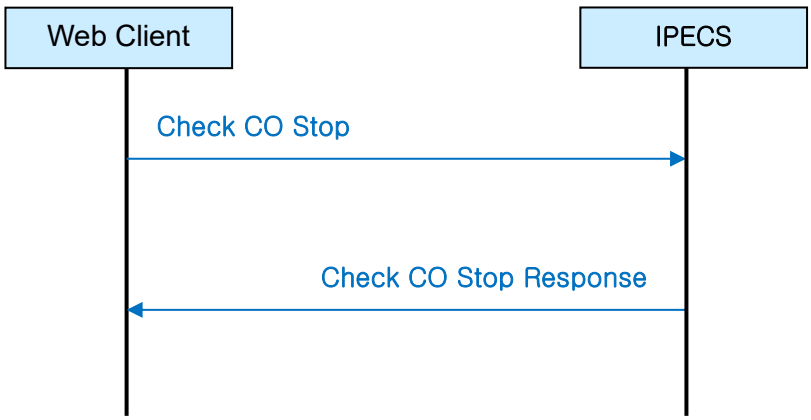
Web Client can send Check CO Stop Request to stop CO status event transmission.

License

API License – CTI required

Usage

Web Client sends a Check CO Stop Request to the iPECS system, and the iPECS system issues a Positive Response if the request command is valid, and a Negative Response if it is not.



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/check_co
Method	**DELETE**
Header	Authorization: Bearer <token from login response>
Parameter	{ "deviceNumber": ["10","11", …]

	}

Positive Response	HTTP 200 OK
	Content:
	{
	“result”:[
	{“10”: ”OK”},
	{“11”: “generic”},
	…
	{“20”: “invalidParameterValue"}
	]
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

The result code is explained in Table 5. Error Code Table.

[Condition]

In case of Negative Response, return Error value as below.

- Operation (serviceNotSupported) - service not provided
- Operation (invalidMonitorObjectType) - invalid type
- Operation (invalidDeviceReferenceIdentifier) - invalid deviceNumber

5.3.4.3. Get Check CO Status

Function Description

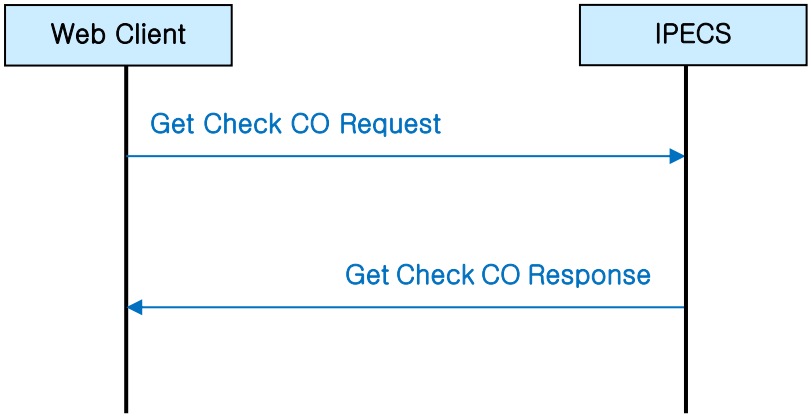
Web Client request to get Check CO status information.

License

API License – CTI required

Usage

Web Client sends Get Check CO Status to iPECS system, and iPECS system issues Positive Response if the request command is valid, and Negative Response if it is not.



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/check_co
Method	GET
Header	Authorization: Bearer <token from login response>
Parameter	• startIndex • count

Positive Response	HTTP 200 OK
	Content:
	{
	"data":{
	"check_cos":[
	{
	"deviceIndex": 1,
	"coNumber": "10",
	"state": "enable"
	},
	…]
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

Request Parameter

Parameter	Value	Mandatory/ Optional	Description
startIndex	Positive integer	M	Terminal physical number
count	Positive integer	M	Terminal number for request, maximum 100

Response Parameter

Parameter	Value	Mandatory/ Optional	Description

deviceIndex	Positive integer	M	CO physical number
coNumber	Digit string	M	CO number
state	String	M	Check CO state, enable/disable

5.3.4.4. Check CO Event

Check CO Event transmits status information for the CO which status has changed.

Parameter	Value	Mandatory/ Optional	Description
type	string	M	check_co
coNumber	Digit string	M	CO number
deviceType	string	M	ext : extension trk : trunk
status	string	M	block idle DND stn_talk trk_talk busy fwd_all fwd_uncond fwd_busy fwd_nans fwd_bsna ring hold conf
evtSerialNum	Positive integer	M	Event serial number

Event:

```
{
   "event":"call",
   "data":{
      "type":"check_co",
      "deviceNumber":"1003",
      "deviceType":"ext",
      "status":"busy",
      "evtSerialNum":1
   }
}
```

[Condition]

The status information of the DN is shown below.

- block: unregistered state
- idle: idle state
- DND: DND setting status
- stn_talk: status of extension call
- trk_talk: Busy state during trunk line

- busy: busy state other than extension / trunk line call
- cfwd_all; Unconditional call forward registration status
- cfwd_busy: call forwarding registration status
- cfwd_nans: No-answer forwarding registration status
- cfwd_bsna: busy / no-answer forwarding registration status
- ring: ringing status
- hold: holding the counter part
- conf: conference status

5.3.5. Call Control Command

Web Client can control the Extension connected to the iPECS system.

5.3.5.1. Get Terminal Information

Function Description

Web Client request to get terminal information.

License

API License – CTI required

Usage

Web Client sends Get Terminal Information to iPECS system, and iPECS system issues Positive Response if the request command is valid, and Negative Response if it is not.

URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/terminals
Method	GET
Header	Authorization: Bearer <token from login response>
Parameter	• startIndex • count

Positive Response	HTTP 200 OK
	Content:
	{
	"data":{
	"terminals":[
	{
	"deviceIndex":1,
	"deviceNumber":"1000",
	“deviceType":"LIP-9040C",
	"tenantNumber":1,
	"callControl":"enable"
	},
	…]
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

Request Parameter

Parameter	Value	Mandatory/	Description

		Optional	
startIndex	Positive integer	M	Terminal physical number
count	Positive integer	M	Terminal number for request, maximum 100

Response Parameter

Parameter	Value	Mandatory/ Optional	Description
deviceIndex	Positive integer	M	Terminal physical number
deviceNumber	Digit string	M	Terminal number
deviceType	String	M	Terminal model name
tenantNumber	Positive integer	M	Tenant (ICM) number
callControl	String	M	Enable means the terminal has license and can be controlled.

5.3.5.2. Make Call

Function Description

Web Client makes one Extension to call.

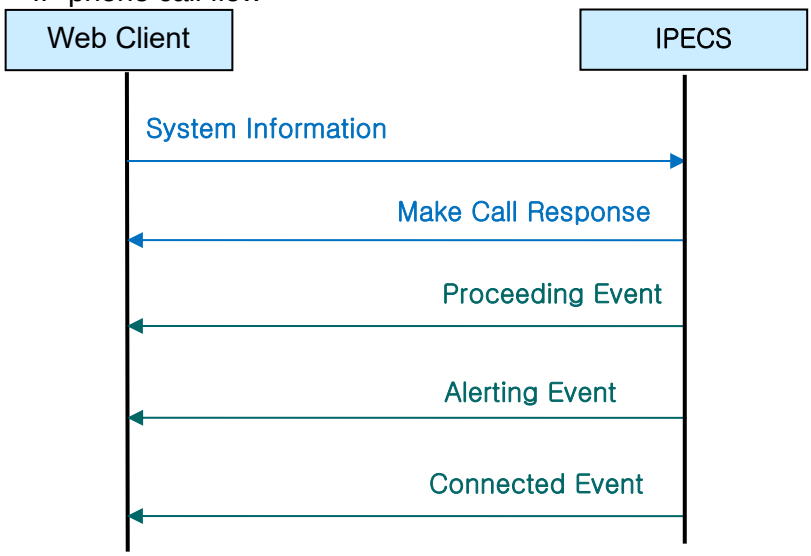
License

API License – CTI required
API License – CTI-EXT required

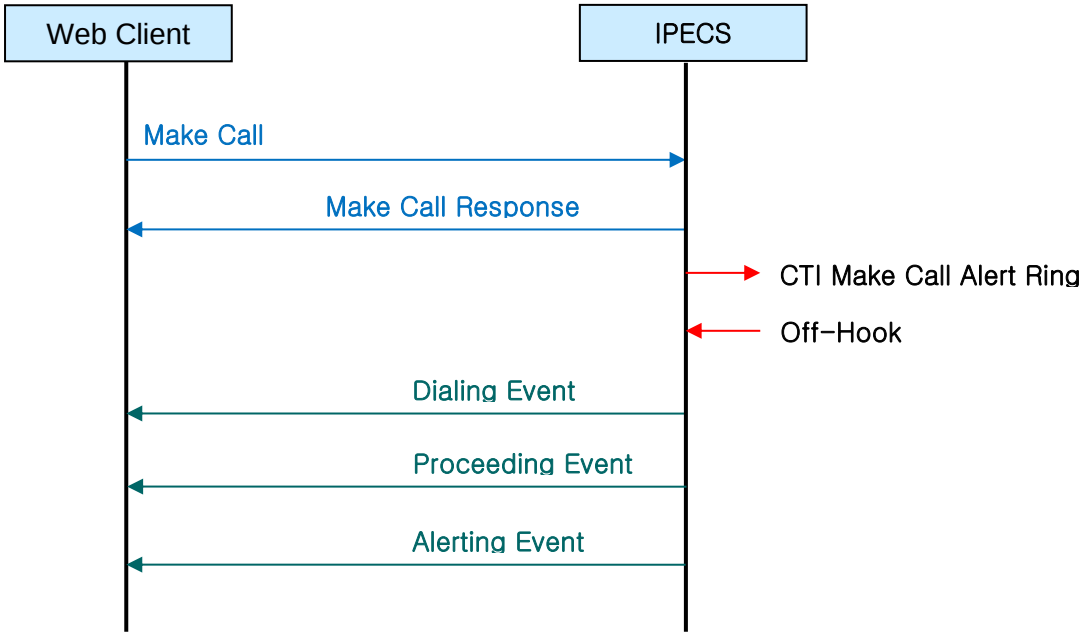
Usage

Web Client sends Make Call to iPECS system, and iPECS system issues Positive Response if the request command is valid, and Negative Response if it is not.

- IP phone call flow



- Analog phone call flow



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/make
Method	POST
Header	Authorization: Bearer <token from login response>
Parameter	• deviceNumber • calledDN

Positive Response	HTTP 200 OK
	Content:
	{
	"data":{
	“callRefID”: 2
	}
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

Parameter	Value	Mandatory/ Optional	Description
deviceNumber	Digit string	M	Terminal number
calledDN	Digit string	M	Called number

[Condition]

For Positive Response,

- Proceeding/Alerting/Connected Event is generated.

[Limitation]

- Possible calledDirectoryNumber : whole number except most of feature code
- At feature code, Group Pickup and paging code can be accepted

5.3.5.3. Answer Call

Function Description

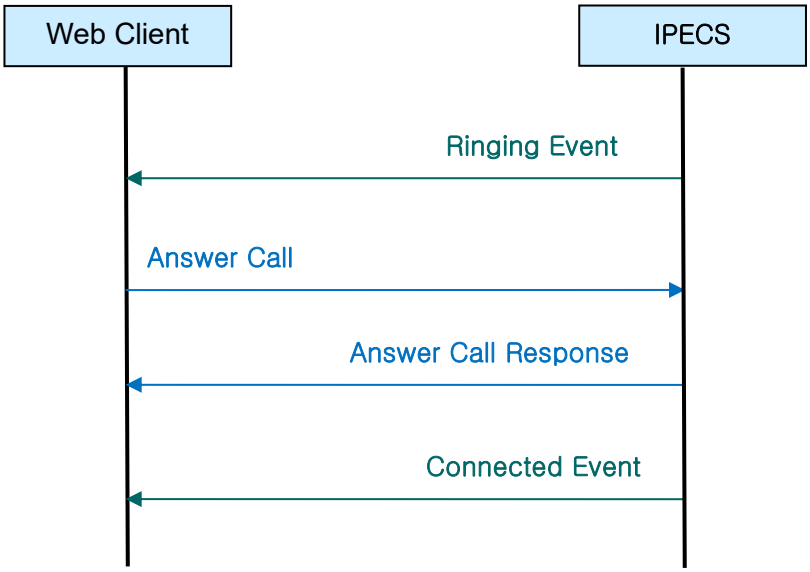
Web Client detects incoming call and makes Extension to answer call

License

API License – CTI required
API License – CTI-EXT required

Usage

Web Client sends Answer Call to iPECS system, and iPECS system issues Positive Response if the request command is valid, and Negative Response if it is not.



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/answer
Method	**POST**
Header	Authorization: Bearer <token from login response>
Parameter	• deviceNumber • callRefID

Positive Response	HTTP 200 OK
	Content:
	{
	“data”: {

	} }

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

Parameter	Value	Mandatory/ Optional	Description
deviceNumber	Digit string	M	Terminal number
callRefID	Positive integer	M	call reference identifier

[Condition]
For Positive Response, Connected Event is generated.

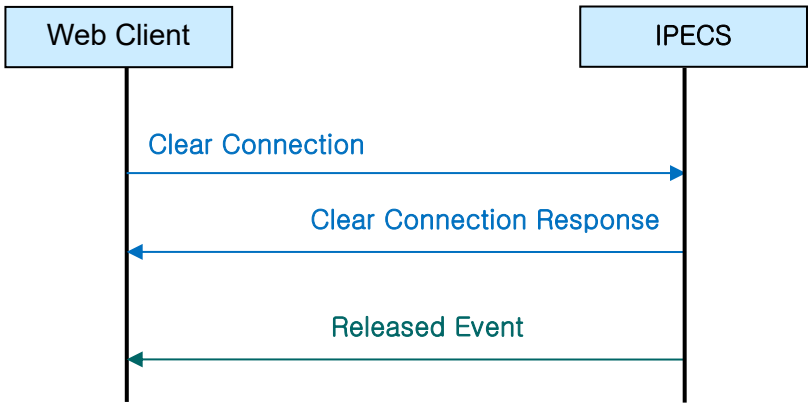
5.3.5.4. Clear Connection

Function Description
Web Client clears call.

License
API License – CTI required
API License – CTI-EXT required

Usage

Web Client sends Clear Connection to iPECS system, and iPECS system issues Positive Response if the request command is valid, and Negative Response if it is not.



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/clear
Method	**POST**
Header	Authorization: Bearer <token from login response>
Parameter	• deviceNumber • callRefID

Positive Response	HTTP 200 OK
	Content:
	{
	“data”: {
	}
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

Parameter	Value	Mandatory/ Optional	Description
deviceNumber	Digit string	M	Terminal number
callRefID	Positive integer	M	Call reference identifier

[Condition]

For Positive Response, Released Event is generated.

For “Negative Responses”, following “Error” values are Returned.

- operation (serviceNotSupported) - unsupported service
- operation (invalidConnectionIdentifier) – Connection ID is not valid.
- stateIncompatibility (noConnectionToClear) – No Connection ID to clear

5.3.5.5. Reroute Call

Function Description

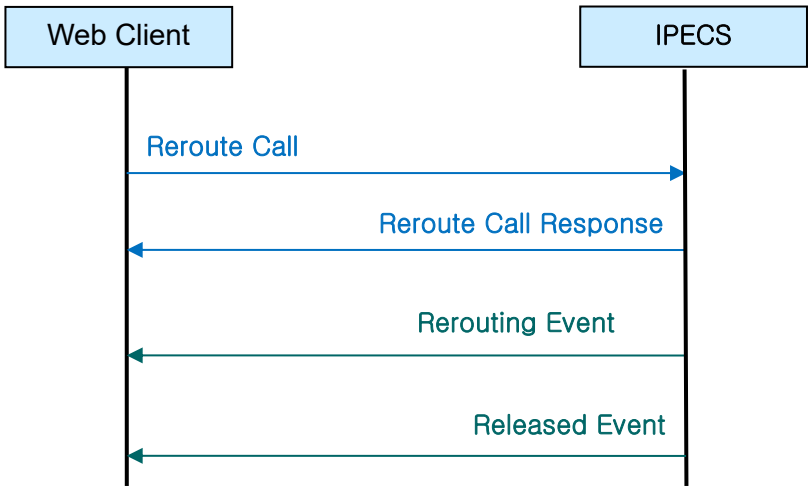
diverts call to other extension, trunk, ACD Group ... This Request can be sent to ACD Group

License

API License – CTI required
API License – CTI-EXT required

Usage

Web Client sends Reroute Call to iPECS system, and iPECS system issues Positive Response if the request command is valid, and Negative Response if it is not.



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/reroute
Method	**POST**
Header	Authorization: Bearer <token from login response>
Parameter	• deviceNumber • callRefID • divertToDN

Positive Response	HTTP 200 OK
	Content:
	{
	“data”: {
	}
	
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

Parameter	Value	Mandatory/ Optional	Description
deviceNumber	Digit string	M	Terminal number
callRefID	Positive integer	M	call reference identifier
divertToDN	Digit string	M	Reroute destination number

[Condition]

For Positive Response, Rerouted Event is generated.

For “Negative Responses”, following “Error” values are Returned.

- operation (serviceNotSupported) - unsupported service
- operation (invalidConnectionIdentifier) – Connection ID is not valid.
- operation (invalidDestination)
- operation (invalidDestinationDeviceObject) – newDestination is not valid type.
- operation (invalidDivertingDeviceIdentifier) – Diverting Device ID is not valid.

[Limitation]

- Possible callToBeDiverted: Whole Number except Feature code, Conference Room

5.3.5.6. Hold Call

Function Description

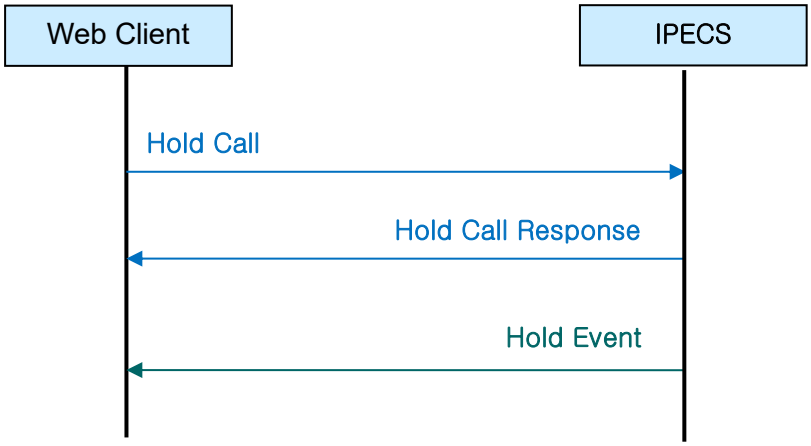
Web Client places call on hold

License

API License – CTI required
API License – CTI-EXT required

Usage

Web Client sends Hold Call to iPECS system, and iPECS system issues Positive Response if the request command is valid, and Negative Response if it is not.



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/hold
Method	**POST**
Header	Authorization: Bearer <token from login response>
Parameter	• deviceNumber • callRefID
Positive Response	HTTP 200 OK Content: {

	“data”: {
	}
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

Parameter	Value	Mandatory/ Optional	Description
deviceNumber	Digit string	M	Terminal number
callRefID	Positive integer	M	Call reference identifier

[Condition]

For Positive Response, Held Event is generated.

For “Negative Responses”, following “Error” values are Returned.

- operation (serviceNotSupported) - unsupported service
- operation (invalidConnectionIdentifier) – Connection ID is not valid.
- operation (invalidHeldDeviceIdentifier)
- stateIncompatibility (invalidConnectionState) – Not calling status
- stateIncompatibility (invalidHeldDeviceState)
- stateIncompatability (NoHeldCall)

5.3.5.7. Retrieve Call

Function Description

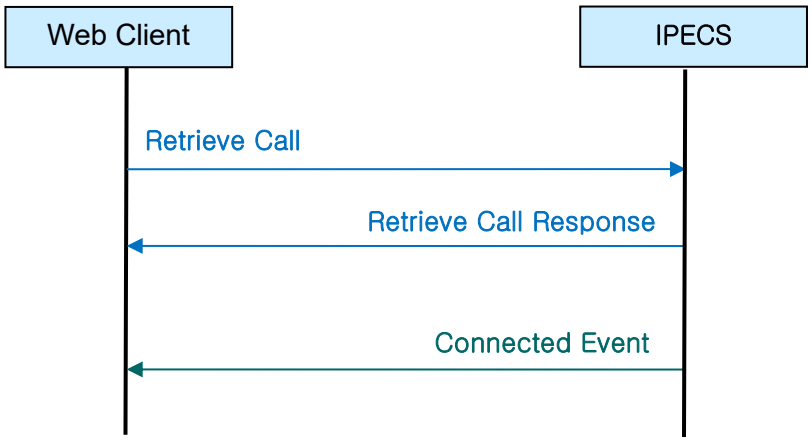
Web Client reconnects held call

License

API License – CTI required
API License – CTI-EXT required

Usage

Web Client sends Retrieve Call to iPECS system, and iPECS system issues Positive Response if the request command is valid, and Negative Response if it is not.



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/retrieve
Method	**POST**
Header	Authorization: Bearer <token from login response>
Parameter	• deviceNumber • callRefID

Positive Response	HTTP 200 OK
	Content:
	{
	“data”: {
	}
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

Parameter	Value	Mandatory/ Optional	Description
deviceNumber	Digit string	M	Terminal number
callRefID	Positive integer	M	Call reference identifier

[Condition]

For Positive Response, Retrieved Event is generated.

For “Negative Responses”, following “Error” values are Returned.

- operation (serviceNotSupported) - unsupported service
- operation (invalidConnectionIdentifier) – Connection ID is not valid.
- operation (invalidHeldDeviceIdentifier)
- stateIncompatibility (invalidConnectionState) – Not calling status.
- stateIncompatibility (invalidHeldDeviceState)
- stateIncompatability (NoHeldCall)

5.3.5.8. Consultation Call

Function Description

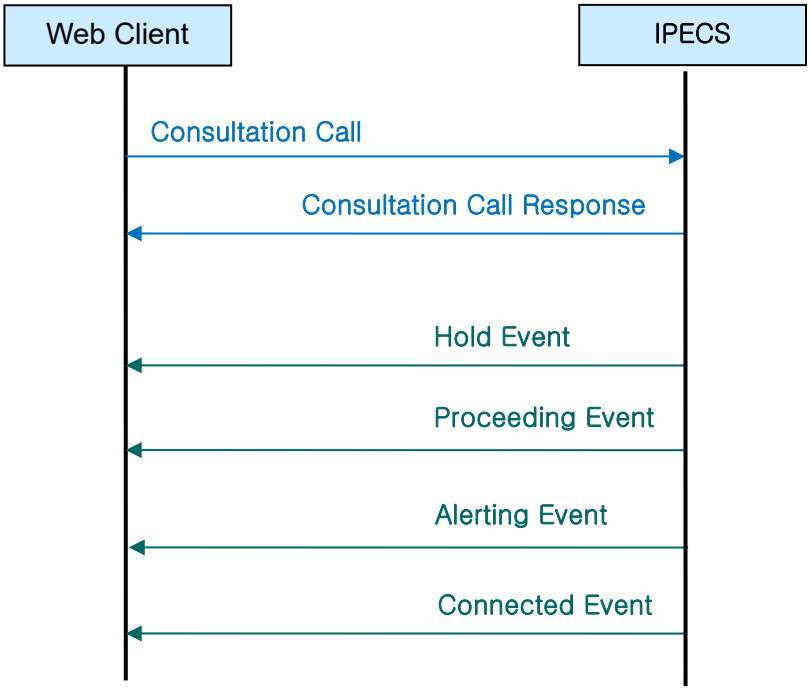
Web Client puts call on hold and starts a new call

License

API License – CTI required
API License – CTI-EXT required

Usage

Web Client sends Consultation Call to iPECS system, and iPECS system issues Positive Response if the request command is valid, and Negative Response if it is not.



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/consultation
Method	**POST**
Header	Authorization: Bearer <token from login response>
Parameter	• deviceNumber • callRefID • calledDN • svcType

Positive Response	HTTP 200 OK
	Content:
	{
	“data”: {
	“callRefID”: 3
	}
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

Parameter	Value	Mandatory/ Optional	Description
deviceNumber	Digit string	M	Terminal number
callRefID	Positive integer	M	Call reference identifier

calledDN	Digit string	M	Called number
svcType	string	M	conference transfer

[Condition]

For Positive Response, Held/Proceeding/Alerting/Connected Event is generated.

For “Negative Responses”, following “Error” values are Returned.

- operation (serviceNotSupported) - unsupported service
- operation (invalidConnectionIdentifier) – Connection ID is not valid.
- operation (invalidDestination)
- operation (invalidDestinationDeviceObject) - newDestination is not valid type.

[Limitation]

- Possible consultedDevice: Whole Number except Feature code, Conference Room

5.3.5.9. Transfer Call

Function Description

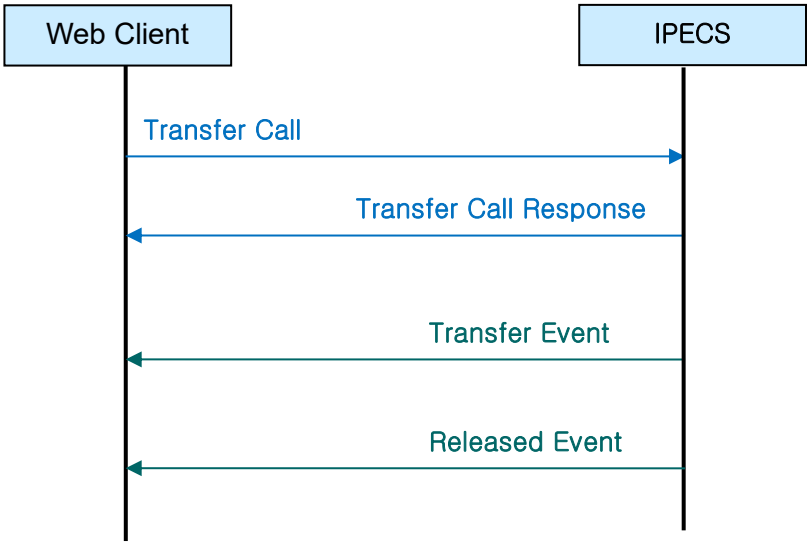
Web Client transfers held call to current active call.
callRefID is reused with that of held call for call tracking.

License

API License – CTI required
API License – CTI-EXT required

Usage

Web Client sends Transfer Call to iPECS system, and iPECS system issues Positive Response if the request command is valid, and Negative Response if it is not.



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/transfer
Method	**POST**
Header	Authorization: Bearer <token from login response>
Parameter	• deviceNumber • callRefID

Positive Response	HTTP 200 OK
	Content:
	{
	“data”: {
	“callRefID”: 3
	}
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

Parameter	Value	Mandatory/ Optional	Description
deviceNumber	Digit string	M	Terminal number
callRefID	Positive integer	M	Call reference identifier

[Condition]

For Positive Response, Transferred Event is generated.

For “Negative Responses”, following “Error” values are Returned.

- operation (serviceNotSupported) - unsupported service
- operation (invalidConnectionIdentifier) – Connection ID is not valid.

- operation (invalidDestination)
- operation (invalidDestinationDeviceObject) - newDestination is not valid type.

5.3.5.10. Sigle Step Transfer Call

Function Description

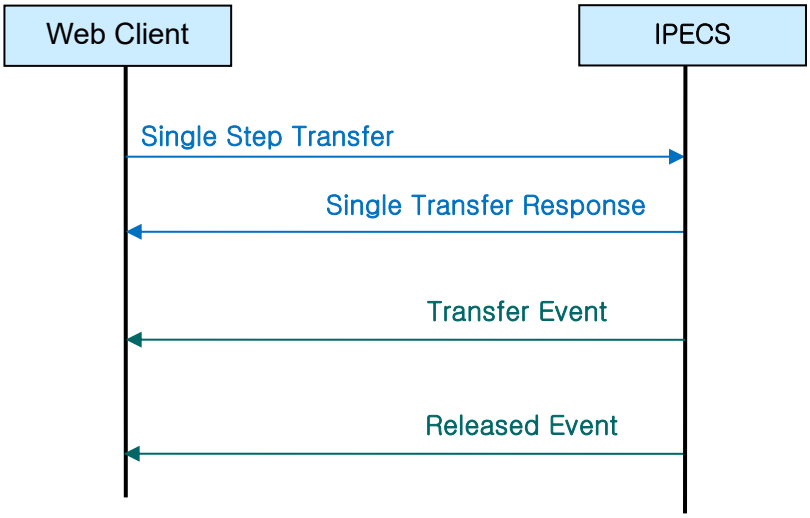
Web Client transfer current call to another number.
This is different from Transfer Call. (No need to put call on hold)

License

API License – CTI required
API License – CTI-EXT required

Usage

Web Client sends Single Step Transfer Call to iPECS system, and iPECS system issues Positive Response if the request command is valid, and Negative Response if it is not.



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/1step_transfer
Method	**POST**
Header	Authorization: Bearer <token from login response>
Parameter	• deviceNumber • callRefID • transferDN

Positive Response	HTTP 200 OK
	Content:
	{
	“data”: {
	“callRefID”: 3
	}

	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

Parameter	Value	Mandatory/ Optional	Description
deviceNumber	Digit string	M	Terminal number
callRefID	Positive integer	M	Call reference identifier
transferDN	Digit string	M	Destination number to transfer

[Condition]

For Positive Response, Transferred Event is generated.

For “Negative Responses”, following “Error” values are Returned.

- operation (serviceNotSupported) - unsupported service
- operation (invalidConnectionIdentifier) – Connection ID is not valid.
- operation (invalidDestination)
- operation (invalidDestinationDeviceObject) - newDestination is not valid type.

[Limitation]

- Possible transferredTo: Whole Number except Feature code, Conference Room

5.3.5.11. Generate Digit(DTMF)

Function Description

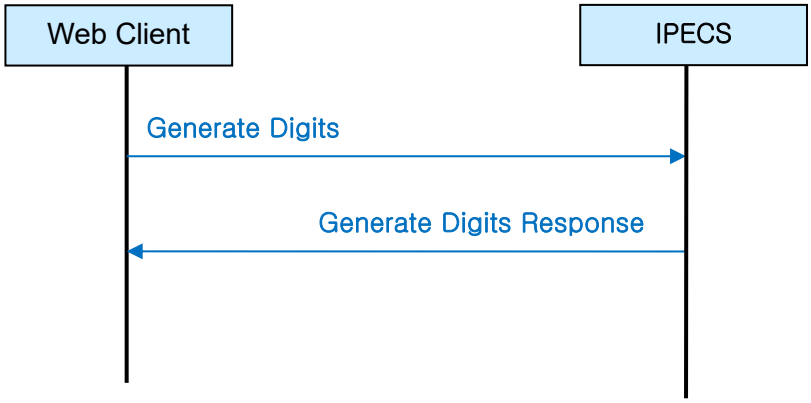
Client sends DTMF digit in talk status.

License

API License – CTI required
API License – CTI-EXT required

Usage

Web Client sends Generate Digit to iPECS system, and iPECS system issues Positive Response if the request command is valid, and Negative Response if it is not.



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/dial_digit
Method	**POST**
Header	Authorization: Bearer <token from login response>
Parameter	• deviceNumber • callRefID • dialDigit

Positive Response	HTTP 200 OK
	Content:
	{
	“data”: {
	}
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

Parameter	Value	Mandatory/ Optional	Description
deviceNumber	Digit string	M	Terminal number
callRefID	Positive integer	M	Call reference identifier
dialDigit	0~9,*,#	M	DTMF digit

[Condition]

For “Negative Responses”, following “Error” values are Returned.

- operation (serviceNotSupported) - unsupported service
- operation (invalidConnectionIdentifier) – Connection ID is not valid.
- operation (invalidParameterValue) – DTMF is not valid.
- stateIncompatibility (invalidConnectionState) – Not calling status.

5.3.5.12. Reconnect Call

Function Description

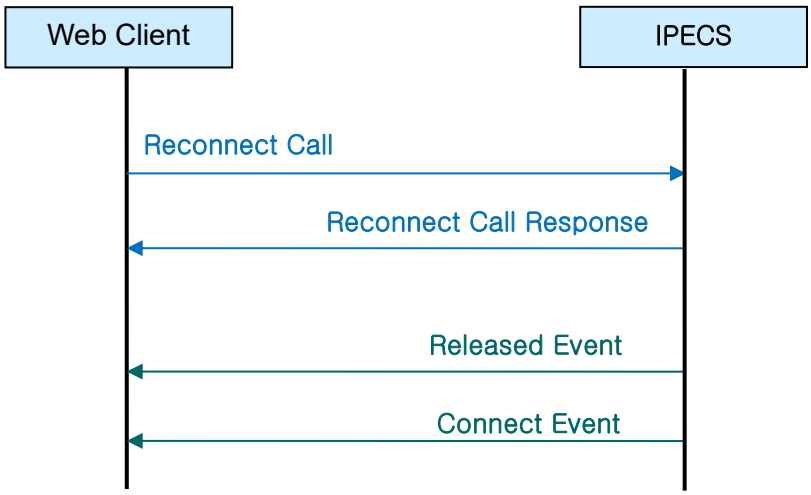
Web Client clears current call and reconnects to held call

License

API License – CTI required
API License – CTI-EXT required

Usage

Web Client sends Reconnect Call to iPECS system, and iPECS system issues Positive Response if the request command is valid, and Negative Response if it is not.



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/reconnect
Method	**POST**
Header	Authorization: Bearer <token from login response>
Parameter	• deviceNumber • callRefID

Positive Response	HTTP 200 OK
	Content:
	{
	“data”: {
	}
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}

	}

Parameter	Value	Mandatory/ Optional	Description
deviceNumber	Digit string	M	Terminal number
callRefID	Positive integer	M	Call reference identifier

[Condition]

For Positive Response, Connection Cleared/Retrieved Event is generated.

For “Negative Responses”, following “Error” values are Returned.

- operation (serviceNotSupported) - unsupported service
- operation (invalidConnectionIdentifier) – Connection ID is not valid.
- operation (invalidHeldDeviceIdentifier)
- stateIncompatibility (invalidConnectionState) – Not calling status.
- stateIncompatibility (invalidHeldDeviceState)
- stateIncompatability (NoHeldCall)

5.3.5.13. Set Forwarding

Function Description

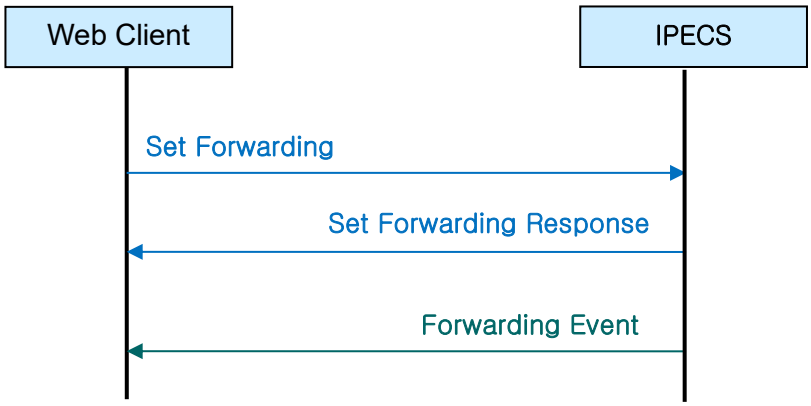
Web Client sets call forwarding.

License

API License – CTI required
API License – CTI-EXT required

Usage

Web Client sends Set Forwarding to iPECS system, and iPECS system issues Positive Response if the request command is valid, and Negative Response if it is not.



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/forward
Method	**POST**
Header	Authorization: Bearer <token from login response>

Parameter	• deviceNumber • DN • cfwdType • destNumber

Positive Response	HTTP 200 OK
	Content:
	{
	“data”: {
	}
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

Parameter	Value	Mandatory/ Optional	Description
deviceNumber	Digit string	M	Terminal number
	Digit string	M	Extension number to register call forwarding
cfwdType	string	M	disable uncondition busy no_answer busy_noans DND
destNumber	Digit string	M	Call forward destination

[Condition]

For Positive Response, Forwarding Event is generated.

For “Negative Responses”, following “Error” values are Returned.

- operation (serviceNotSupported) - unsupported service
- operation (invalidForwardingDestination) – Forwarding Destination is not valid.

5.3.5.14. Get Forwarding

Function Description

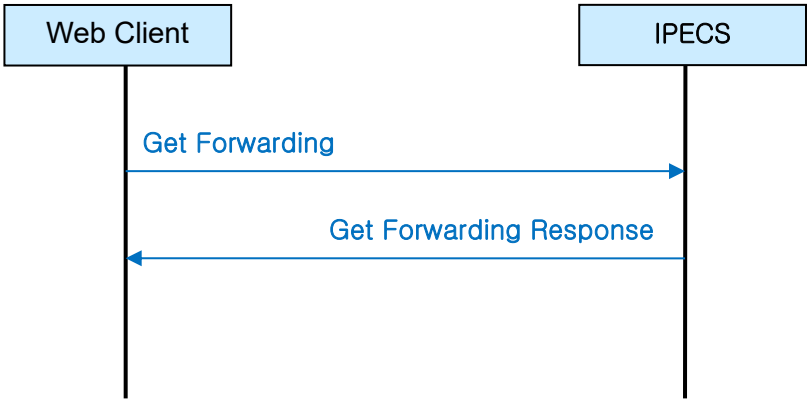
Web Client gets current forwarding status

License

API License – CTI required
API License – CTI-EXT required

Usage

Web Client sends Get Forwarding to iPECS system, and iPECS system issues Positive Response if the request command is valid, and Negative Response if it is not.



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/forward
Method	**GET**
Header	Authorization: Bearer <token from login response>
Parameter	• deviceNumber • DN

Positive Response	HTTP 200 OK
	Content:
	{
	“data”:{
	“cfwdType”: “uncondition”,
	“destNumber”: “901012345678”,
	“ringTime”: 10
	}
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

Request Parameter

Parameter	Value	Mandatory/ Optional	Description
deviceNumber	Digit string	M	Terminal number
DN	Digit string	M	Extension number to get call forwarding

Response Parameter

Parameter	Value	Mandatory/ Optional	Description

cfwdType	string	M	disable uncondition busy no_answer busy_noans DND
destNumber	Digit string	M	Call forwarding destination
ringTime	Int	O	No answer ring time

5.3.5.15. Set DND

Function Description

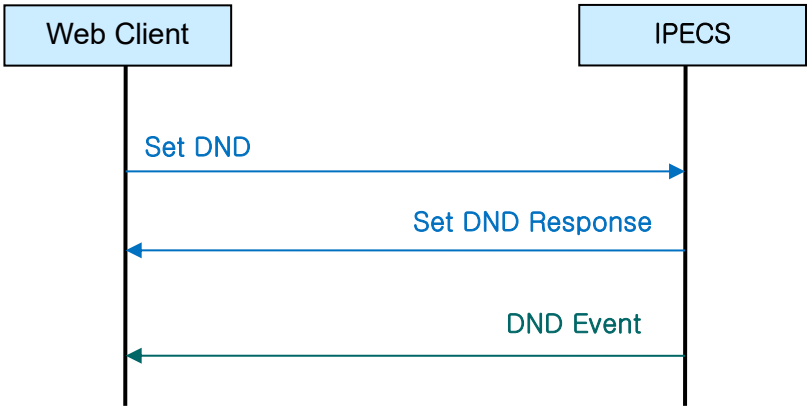
Web Client sets DND for Extension

License

API License – CTI required
API License – CTI-EXT required

Usage

Web Client sends Set DND to iPECS system, and iPECS system issues Positive Response if the request command is valid, and Negative Response if it is not.



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/dnd
Method	**POST**
Header	Authorization: Bearer <token from login response>
Parameter	• deviceNumber • DN • svcType

Positive Response	HTTP 200 OK
	Content:
	{
	“data”: {
	}
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

Parameter	Value	Mandatory/ Optional	Description
deviceNumber	Digit string	M	Terminal number
DN	Digit string	M	Extension number to set DND
svcType	string	M	disable enable

[Condition]

For Positive Response, DND Event is generated.

For "Negative Responses", following "Error" values are Returned.

- operation (serviceNotSupported) - unsupported service

5.3.5.16. Get DND

Function Description

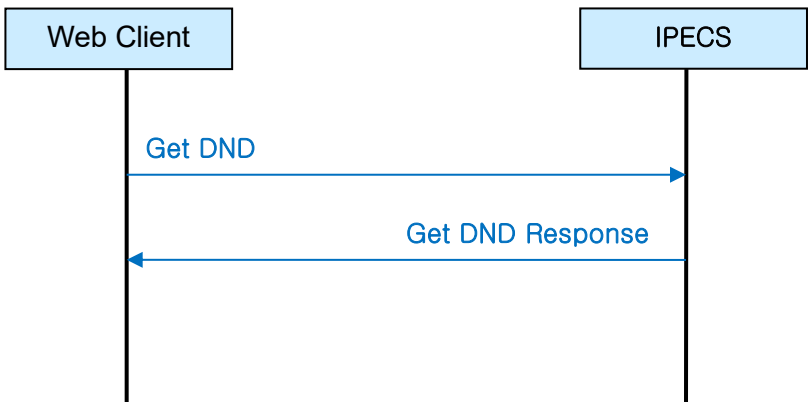
Web Client gets current DND status

License

API License – CTI required
API License – CTI-EXT required

Usage

Web Client sends Get DND to iPECS system, and iPECS system issues Positive Response if the request command is valid, and Negative Response if it is not.



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/dnd
Method	**GET**
Header	Authorization: Bearer <token from login response>
Parameter	• deviceNumber • DN

Positive Response	HTTP 200 OK
	Content:
	{
	“data”:{
	“svcType”: “enable”
	}
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

Request Parameter

Parameter	Value	Mandatory/ Optional	Description
deviceNumber	Digit string	M	Terminal number
DN	Digit string	M	Extension number to get DND status

Response Parameter

Parameter	Value	Mandatory/ Optional	Description
svcType	String	M	disable enable

5.3.5.17. PickUp

Function Description

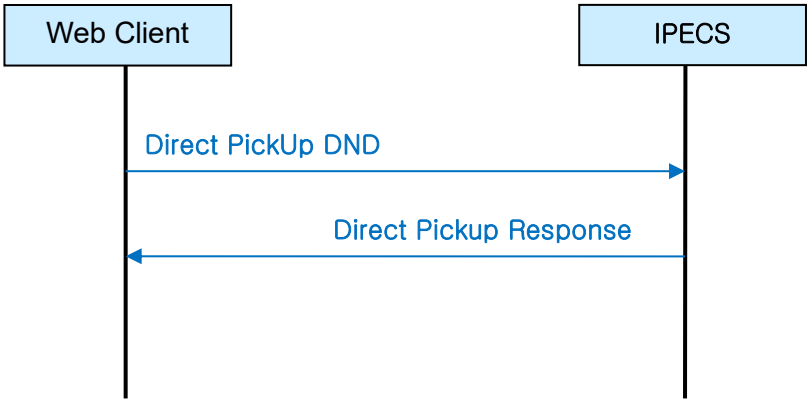
Web Client makes Extension to pickup calls to other phones

License

API License – CTI required
API License – CTI-EXT required

Usage

Web Client sends Direct Pickup to iPECS system, and iPECS system issues Positive Response if the request command is valid, and Negative Response if it is not.



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/pickup
Method	**POST**
Header	Authorization: Bearer <token from login response>
Parameter	• deviceNumber • callingDN • pickedDN • pickupType

Positive Response	HTTP 200 OK
	Content:
	{
	“data”: {
	}
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

Parameter	Value	Mandatory/ Optional	Description
deviceNumber	Digit string	M	Terminal number
callingDN	Digit string	M	Picking-up extension number
pickedDN	Digit string	O	Pick-up destination number
pickupType	string	M	group_pickup direct_pickup

[Condition]

For “Negative Responses”, following “Error” values are Returned.

- operation (serviceNotSupported) - unsupported service
- operation (invalidConnectionIdentifier) – Connection ID is not valid.
- operation (invalidDestination)

- operation (invalidDestinationDeviceObject) – newDestination is not valid type.
- stateIncompatibility (invalidConnectionState) – Not Idle status
- stateIncompatibility (generic)

[Limitation]

- Possible callToBePickedUp: Only ringing device can be picked up

5.3.5.18. Conference Call

Function Description

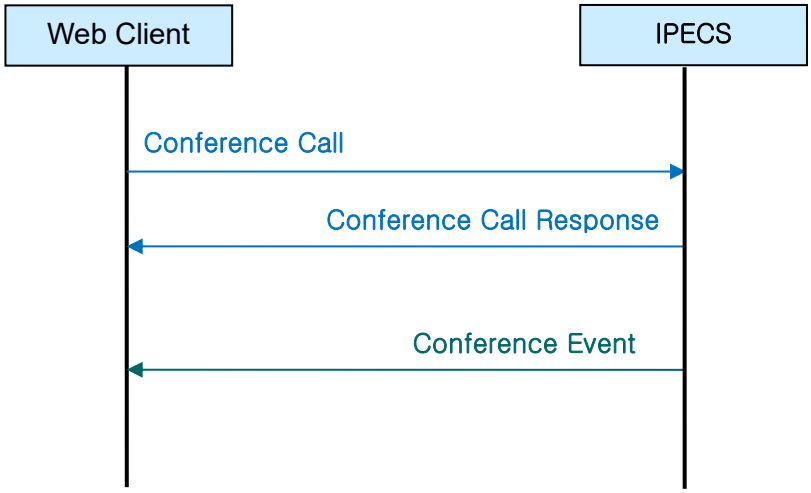
Web Client can make a conference call with the held call and the active call.

License

API License – CTI required
API License – CTI-EXT required

Usage

Web Client sends Conference Call to iPECS system, and iPECS system issues Positive Response if the request command is valid, and Negative Response if it is not.



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/conference
Method	**POST**
Header	Authorization: Bearer <token from login response>
Parameter	• deviceNumber • callRefID
Positive Response	HTTP 200 OK Content:

	{
	“data”: {
	“callRefID”: 3
	}
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

Parameter	Value	Mandatory/ Optional	Description
deviceNumber	Digit string	M	Terminal number
callRefID	Positive integer	M	Call reference identifier

[Condition]

For Positive Response, Connection Cleared Event is generated.

For “Negative Responses”, following “Error” values are Returned.

- operation (serviceNotSupported) - unsupported service
- operation (invalidConnectionIdentifier) – Connection ID is not valid.
- operation (invalidConnectionState) – Connection State is not valid

5.3.5.19. Call Reject

Function Description

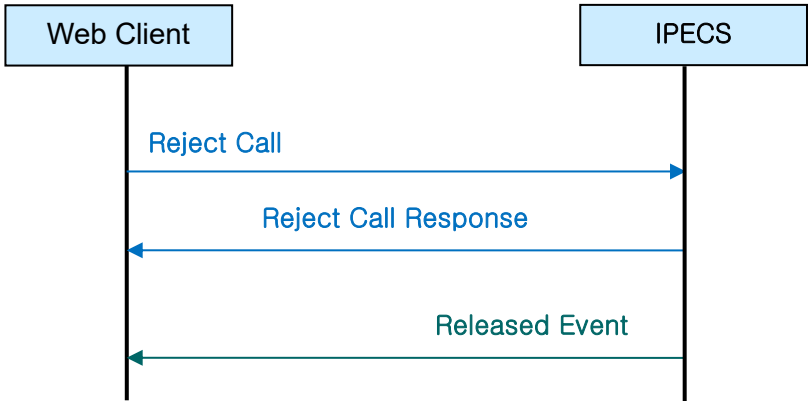
Web Client can reject a phone call at ring state

License

API License – CTI required
API License – CTI-EXT required

Usage

Send Reject Call Request message when Extension is ring state



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/reject
Method	**POST**
Header	Authorization: Bearer <token from login response>
Parameter	• deviceNumber • callRefID

Positive Response	HTTP 200 OK
	Content:
	{
	“data”: {
	}
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

Parameter	Value	Mandatory/ Optional	Description
deviceNumber	Digit string	M	Terminal number
callRefID	Positive integer	M	Call reference identifier

[Condition]

For Positive Response, Connection Cleared Event is generated.

For “Negative Responses”, following “Error” values are Returned.

- operation (serviceNotSupported) - unsupported service
- operation (invalidConnectionIdentifier) – Connection ID is not valid.
- stateIncompatibility (noConnectionToClear) – No Connection ID to clear

5.3.5.20. BusyService

Function Description

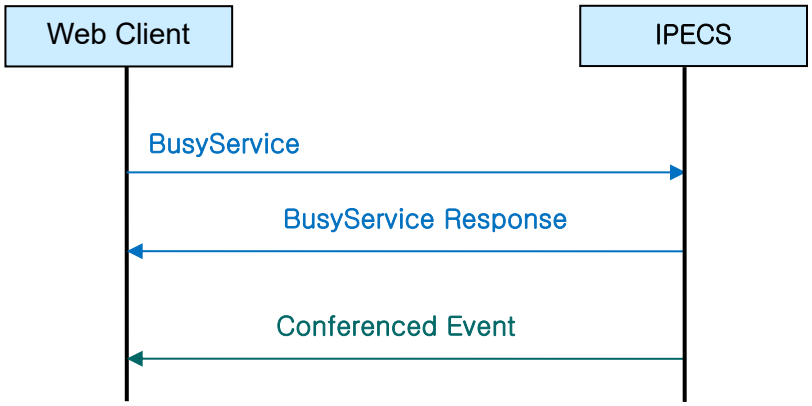
Web Client can request supplementary service in destination busy state.

License

API License – CTI required
API License – CTI-EXT required

Usage

Send BusyService Request message to the system when you hear busy tone.



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/busy_service
Method	**POST**
Header	Authorization: Bearer <token from login response>
Parameter	• deviceNumber • callRefID • type

Positive Response	HTTP 200 OK
	Content:
	{
	“data”: {
	“callRefID”: 3
	}
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

Parameter	Value	Mandatory/ Optional	Description
deviceNumber	Digit string	M	Terminal number
callRefID	Positive integer	M	Call reference identifier
type	string	M	[Busy-service list] intrude callback campon callwait

[Condition]

This request only operated at busy tone state which is just after failed make call.

For “Negative Responses”, following “Error” values are Returned.

- operation (serviceNotSupported) - unsupported service
- operation (accessDeny) – Check intrusion allowed at iPECS SME
- operation (invalidState) – Phone state is wrong

5.3.5.21. Set Call Back

Function Description

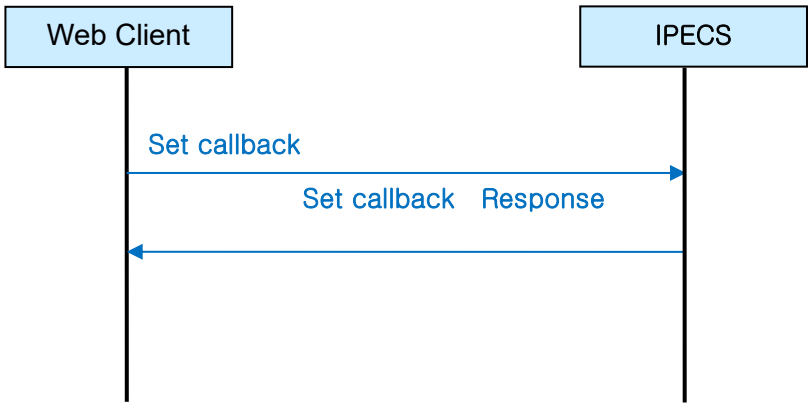
Cancel CallBack

License

API License – CTI required
API License – CTI-EXT required

Usage

Send Set CallBack message to the system.



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/call_back
Method	**POST**
Header	Authorization: Bearer <token from login response>
Parameter	• deviceNumber • svcType

Positive Response	HTTP 200 OK
	Content:
	{
	“data”: {
	}
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

Parameter	Value	Mandatory/ Optional	Description
deviceNumber	Digit string	M	Terminal number
svcType	string	M	false

[Condition]

This request only operated at busy tone state which is just after failed make call.

For “Negative Responses”, following “Error” values are Returned.

- operation (serviceNotSupported) - unsupported service
- operation (accessDeny) – Check intrusion allowed at iPECS SME
- operation (invalidState) – Phone state is wrong

5.3.5.22. Get Call Status

Function Description

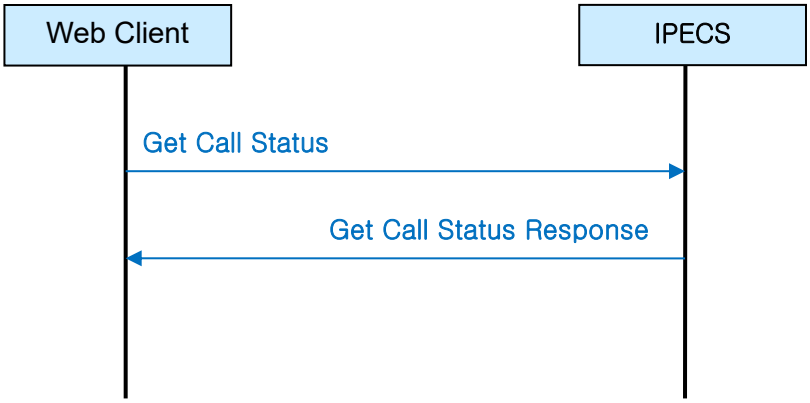
Web Client requests the call status of a specific extension.

License

API License – CTI required
API License – CTI-EXT required

Usage

Web Client sends Get Call Status to iPECS system, and iPECS system issues Positive Response if the request command is valid, and Negative Response if it is not.



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/status
Method	**GET**
Header	Authorization: Bearer <token from login response>
Parameter	• deviceNumber

Positive Response	HTTP 200 OK
	Content:
	{
	“data”: {
	“state”: “talk”
	}
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

Request Parameter

Parameter	Value	Mandatory/ Optional	Description
deviceNumber	Digit string	M	Terminal number

Response Parameter

Parameter	Value	Mandatory/ Optional	Description
State	block idle DND stn_talk trk_talk busy fwd_all fwd_busy	M	Same with presence status

	fwd_nans fwd_bsna ring hold conf		

5.3.5.23. Call Park

Function Description

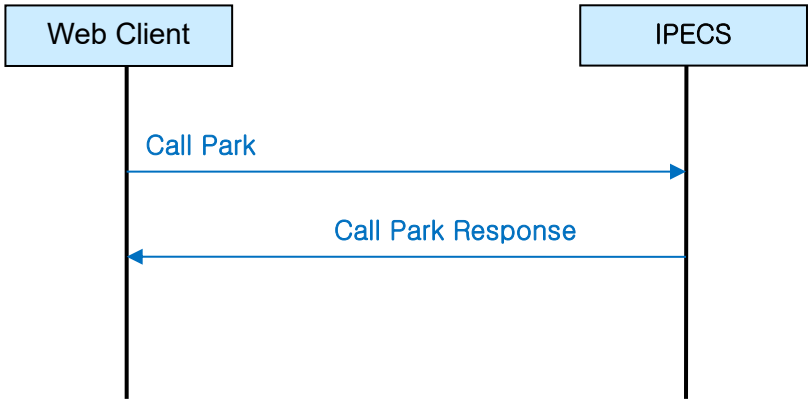
This function is to register call-park

License

API License – CTI required
API License – CTI-EXT required

Usage

Web Client sends Call Park to iPECS system, and iPECS system issues Positive Response if the request command is valid, and Negative Response if it is not.



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/park
Method	**POST**
Header	Authorization: Bearer <token from login response>
Parameter	• deviceNumber • callRefID

Positive Response	HTTP 200 OK
	Content:
	{
	“data”: {
	“parkingLoc”: 1
	}
	}

Negative Response	HTTP 401 Unauthorized, HTTP 403 Forbidden HTTP 404 Not Found Content: {

	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

Parameter	Value	Mandatory/ Optional	Description
deviceNumber	Digit string	M	Terminal number
callRefID	Positive integer	M	callRefID to call-park

[Condition]

For “Negative Responses”, following “Error” values are Returned.

- operation (serviceNotSupported) - unsupported service
- operation (invalidConnectionIdentifier) – invalid Connection ID
- operation (invalidDestination)
- stateIncompatibility (generic)

5.3.5.24. Call UnPark

Function Description

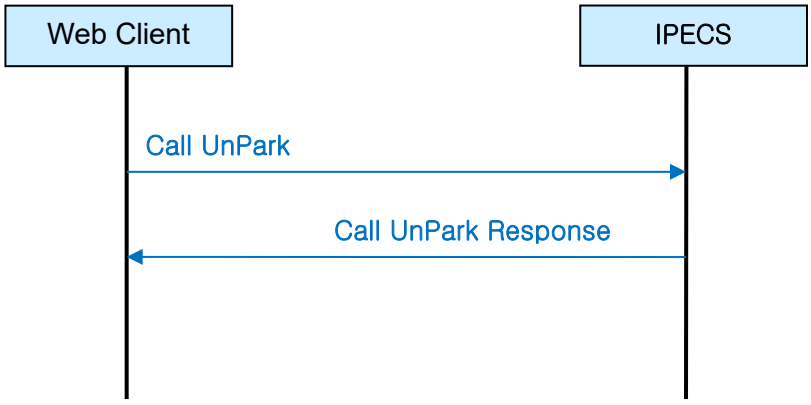
This function is to retrieve parked call

License

API License – CTI required
API License – CTI-EXT required

Usage

Web Client sends Call UnPark to iPECS system, and iPECS system issues Positive Response if the request command is valid, and Negative Response if it is not.



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/unpark
Method	**POST**
Header	Authorization: Bearer <token from login response>

Parameter	• deviceNumber • callingDN • parkingLoc

Positive Response	HTTP 200 OK
	Content:
	{
	“data”: {
	}
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

Parameter	Value	Mandatory/ Optional	Description
deviceNumber	Digit string	M	Terminal number
callingDN	Digit string	M	DN retrieving parked-call
parkingLoc	int	M	Parking location range (000~999)

[Condition]

For “Negative Responses”, following “Error” values are Returned.

- operation (serviceNotSupported) - unsupported service
- operation (invalidConnectionIdentifier) – invalid Connection ID
- operation (invalidDestination)
- stateIncompatibility (generic)

5.3.5.25. Get MWI (Message Waiting Indicator)

Function Description

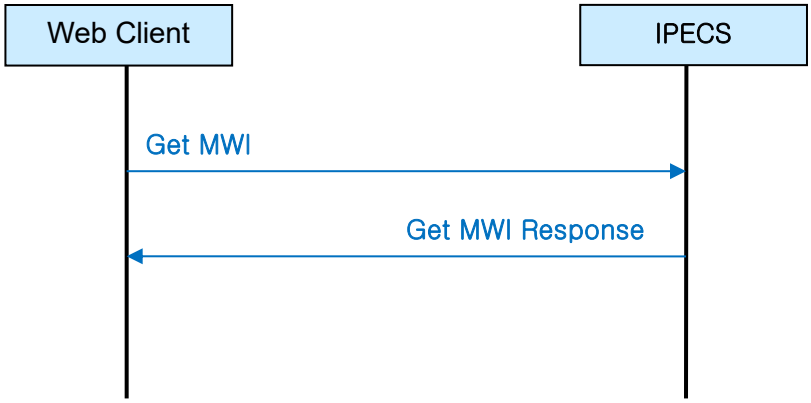
This function is to get message waiting status

License

API License – CTI required
API License – CTI-EXT required

Usage

Web Client sends Get MWI to iPECS system, and iPECS system issues Positive Response if the request command is valid, and Negative Response if it is not.



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/mwi
Method	**GET**
Header	Authorization: Bearer <token from login response>
Parameter	• deviceNumber

Positive Response	HTTP 200 OK
	Content:
	{
	“data”:{
	“msgStatus”: “true”,
	“newMsg”:34,
	“oldMsg”:12,
	“newUrgentMsg”: 56
	}
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

Request Parameter

Parameter	Value	Mandatory/ Optional	Description
deviceNumber	Digit string	M	Terminal number

Response Parameter

Parameter	Value	Mandatory/ Optional	Description
msgStatus	string	M	true : new message exists false :new message doesn’t exist
newMsg	Positive integer	M	New message count
oldMsg	Positive integer	M	Old message count

newUrgentMsg	Positive integer	M	New urgent message count

5.3.5.26. Flexible Page

Function Description

Request Flexible Page according to predefined program

License

API License – CTI required

Usage

The system administrator must program “Flexible Page Feature” menu first.
Client sends Flexible Page request with index of “Flexible Page Feature” menu to iPECS system, and iPECS system issues Positive Response if the request command is valid, and Negative Response if it is not.

Program related with paging.

(1) System Data – Flexible Page Feature

Order	Name Max 32 Characters	Page Zone INT(1-100)/INT_ALL(101) EXT1(102)/EXT2(103)/EXT_ALL(104) INT_EXT_ALL(105) SYS_ALL(116)	Action	Annc.No (0,1-200)	Silent Text No. (0, 1-60)	SMS Only	Page Urgency Level (0-255, 0>1>2..>255)
1	Live_Alarm	2	Live Voice ▼	0	0	OFF ▼	0
2	Annc_Alarm	2	Announce Play ▼	70	2	OFF ▼	0
3	Text_Alarm	2	Silent Message ▼	1	1	OFF ▼	0
4	Annc_Alarm2	3	Announce Play ▼	70	0	OFF ▼	0

(2) System Data – Silent Text Messages

Index		Value	Range
1	Subject	Text Test	Max 32 Characters
	Contents	Help Me	Max 80 Characters
2	Subject	Text with Annc	Max 32 Characters
	Contents	Text with Annc	Max 80 Characters

URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/flexpage
Method	**POST**
Header	Authorization: Bearer <token from login response>

Parameter	• page_index • service_num

Positive Response	HTTP 200 OK
	Content:
	{
	“data”:{
	}
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

Parameter	Value	Mandatory/ Optional	Description
page_index	Unsigned integer	M	Flexible Page index (order number) in Flexible Page Feature menu
service_num	Digit string	M	extension number related with page

5.3.5.27. ACD Agent LogOn

Function Description

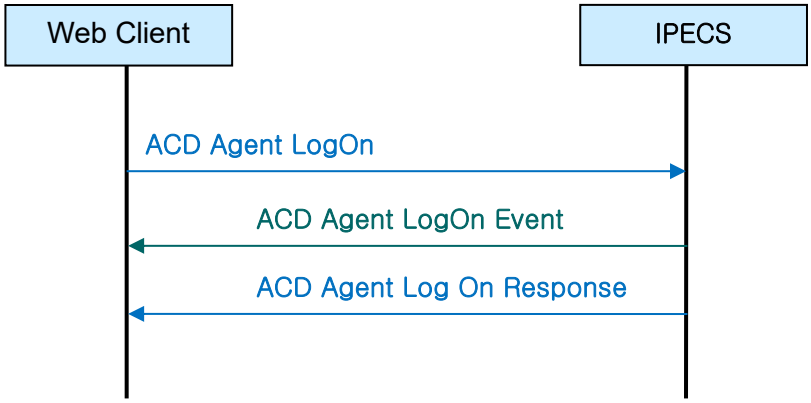
Agent can log on to ACD group by Web API

License

API License – CTI required
API License – CTI-EXT required

Usage

Web Client sends ACD Agent LogOn to iPECS system, and iPECS system issues ACD Agent LogOn Event and Positive Response if the request command is valid, and Negative Response if it is not or Agent has already logged on state.



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/agent_logon
Method	**POST**
Header	Authorization: Bearer <token from login response>
Parameter	• deviceNumber • agentId • groupNumber • loginPrimary • loginPriority

Positive Response	HTTP 200 OK
	Content:
	{
	“data”: {
	}
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

Parameter	Value	Mandatory/ Optional	Description
deviceNumber	Digit string	M	Terminal number
agentId	Digit string	M	Agent ID for log in, 4 digits (0~9)
groupNumber	Digit string	M	ACD Group Number
loginPrimary	string	M	primary secondary
loginPriority	Unsigned integer	M	0~9

5.3.5.28. ACD Agent LogOff

Function Description

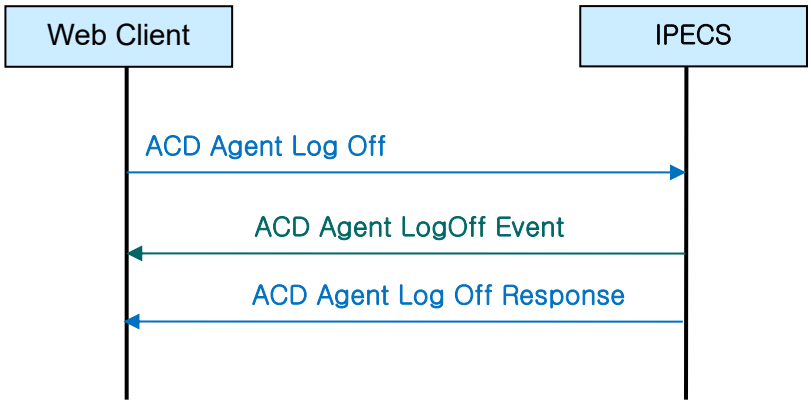
Agent can log off to ACD group by Web API

License

API License – CTI required
API License – CTI-EXT required

Usage

Web Client sends Agent LogOff to iPECS system, and iPECS system issues ACD Agent LogOff Event and Positive Response if the request command is valid, and Negative Response if it is not or Agent is not logged on state.



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/agent_logoff
Method	**POST**
Header	Authorization: Bearer <token from login response>
Parameter	• deviceNumber • agentId • groupNumber • loginPrimary

Positive Response	HTTP 200 OK
	Content:
	{
	“data”: {
	}
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

Parameter	Value	Mandatory/ Optional	Description
deviceNumber	Digit string	M	Terminal number
agentId	Digit string	M	Agent ID for log in, 4 digits (0~9)
groupNumber	Digit string	M	ACD Group Number

loginPrimary	string	M	primary secondary

5.3.5.29. ACD Agent Set State

Function Description

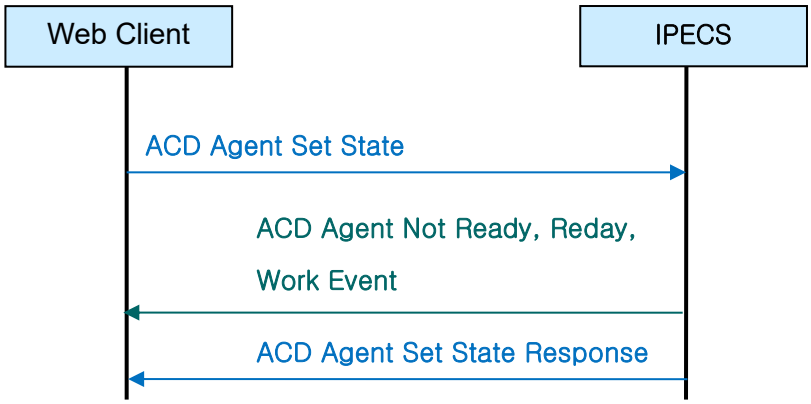
Agent can change it's state by Web API,

License

API License – CTI required
API License – CTI-EXT required

Usage

Web Client sends ACD Agent Set State to iPECS system, and iPECS system issues ACD Agent Not Ready, Ready, Work Event and Positive Response if the request command is valid, and Negative Response if it is not or Agent is not logged on state.



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/agent_state
Method	**POST**
Header	Authorization: Bearer <token from login response>
Parameter	• deviceNumber • agentId • groupNumber • agentState • reasonCode

Positive Response	HTTP 200 OK
	Content:
	{
	"data": {
	}
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {

	"code": xxx, "message": "xx xxxx" } }

Parameter	Value	Mandatory/ Optional	Description
deviceNumber	Digit string	M	Terminal number
agentId	Digit string	M	Agent ID for log in, 4 digits (0~9)
groupNumber	Digit string	M	ACD Group Number
agentState	string	M	Choose below string dutyOn dutyOff
reasonCode	Digit string	M	0~9, *. #

5.3.5.30. ACD Group Control

Function Description

Web API can control ACD Group by Routing queued call to Agent by using “Reroute Call” API. Refer to “Reroute Call”

5.3.5.31. Conference Group Call Start

Function Description

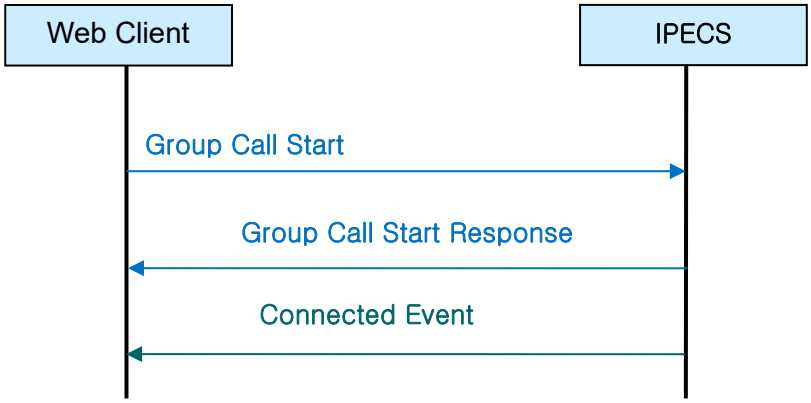
Web Client assigns a specific member (extension, trunk line) terminals and make a conference group call.

License

API License – CTI required

Usage

Web Client sends Conference Group Call Start Argument to iPECS system and iPECS system issues Positive Response if the request command is valid, and Negative Response if it is not.



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/confgroup
Method	POST
Header	Authorization: Bearer <token from login response>
Parameter	• deviceNumber • groupIndex

Positive Response	HTTP 200 OK
	Content:
	{
	“data”: {
	}
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

Parameter	Value	Mandatory/ Optional	Description
deviceNumber	Digit string	M	Device Number
groupIndex	Positive integer	M	Station or System conference group index(1~99 : station group/ 100~265 : system group / 999: cancel)

[**Condition**]
If Positive Response, Proceeding/Alerting/Connected Event will be issued. Alerting will be issued by Caller + Callee Count

[**restriction**]
- possible called Number: Whole type of number without feature code

5.3.5.32. Conference Group Call Add Callee

Function Description

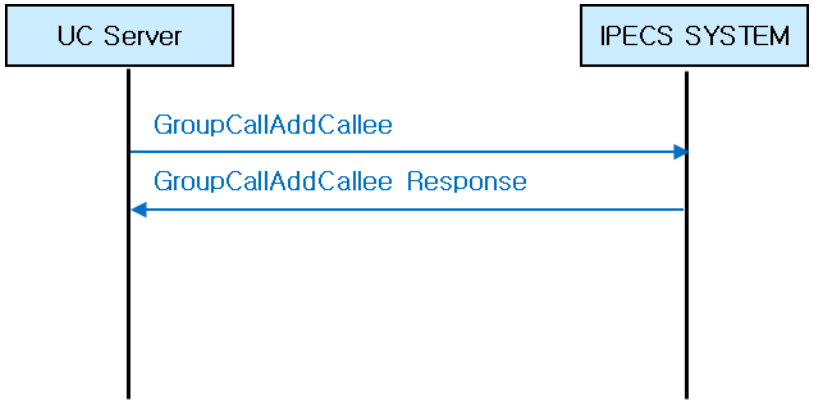
Add Specific Member(Extension, Trunk) during Conference Group Call.

License

API License – CTI required

Usage

- Web Client sends Conference Group Call Add Callee Argument to iPECS system, and iPECS system issues Positive
  Response if the request command is valid, and Negative Response if it is not.
-



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/confgroup_callee
Method	POST
Header	Authorization: Bearer <token from login response>
Parameter	• deviceNumber • groupIndex • memberType • memberIndex • memberDN

Positive Response	HTTP 200 OK
	Content:
	{
	“data”: {
	}
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {

	"code": xxx, "message": "xx xxxx" } }

Parameter	Value	Mandatory/ Optional	Description
deviceNumber	Digit string	M	Device Number
groupIndex	Positive integer	M	Station or System conference group index(1~99 : station group/ 100~265 : system group / 999: cancel)
memberType	String	M	Group call member type, station / co / net
memberIndex	Positive integer	M	Member Index on Conference Group
memberDN	Digit string	M	New Member Number

[Condition]

- this request for group call master

[**restriction**]

- possible called Number: Whole type of number without feature code

5.3.5.33. Conference Group Call Drop Callee

Function Description

Add Specific Member(Extension, Trunk) during Conference Group Call.

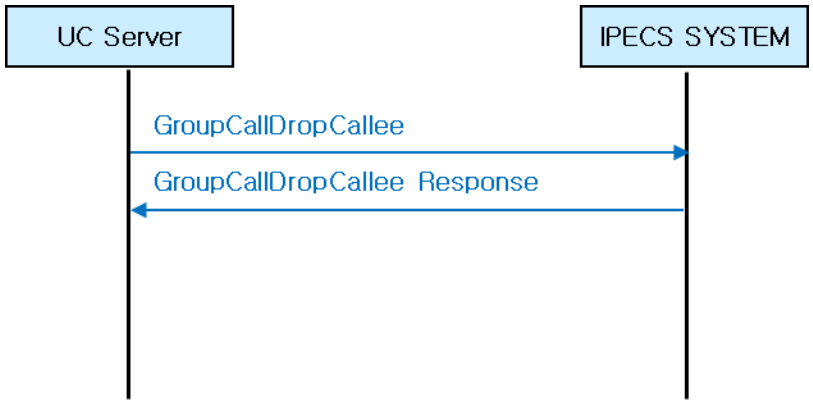
License

API License – CTI required

Usage

- Web Client sends Conference Group Call Drop Callee Argument to iPECS system, and iPECS system issues Positive

  Response if the request command is valid, and Negative Response if it is not.



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/confgroup_callee
Method	DELETE
Header	Authorization: Bearer <token from login response>
Parameter	• deviceNumber • groupIndex • memberIndex

Positive Response	HTTP 200 OK
	Content:
	{
	"data": {
	}
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

Parameter	Value	Mandatory/ Optional	Description
deviceNumber	Digit string	M	Device Number
groupIndex	Positive integer	M	Station or System conference group index(1~99 : station group/ 100~265 : system group / 999: cancel)
memberIndex	Positive integer	M	Member Index on Conference Group

[Condition]

- this request for group call master

[**restriction**]

- possible called Number : DN which in group call

5.3.5.34. Conference Group Call Set Mute On

Function Description

Mute Specific Member(Extension, Trunk) during Group Call.

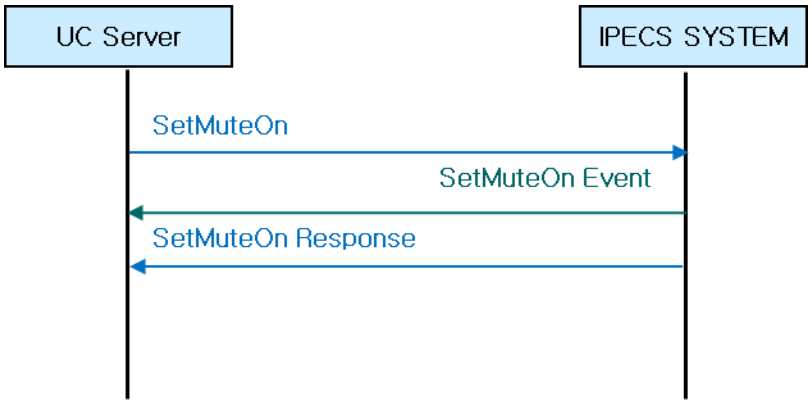
License

API License – CTI required

Usage

- Web Client sends Conference Group Call Drop Callee Argument to iPECS system, and iPECS system issues Positive

  Response if the request command is valid, and Negative Response if it is not.
- Call flow



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/confgroup_mutecallee
Method	POST
Header	Authorization: Bearer <token from login response>
Parameter	• deviceNumber • groupIndex • memberIndex • svcType

Positive Response	HTTP 200 OK
	Content:
	{
	“data”: {
	}
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

Parameter	Value	Mandatory/ Optional	Description
deviceNumber	Digit string	M	Device Number
groupIndex	Positive integer	M	Station or System conference group index(1~99 : station group/ 100~265 : system

			group / 999: cancel)
memberIndex	Positive integer	M	Member Index on Conference Group
svcType	String	M	on or off

[Condition]
- this request for group call master

[Restriction]
- possible called Number : DN which in group call

5.3.5.35. Get conference room state

Function Description

Web Client can get Conference Room state which is use or not use.

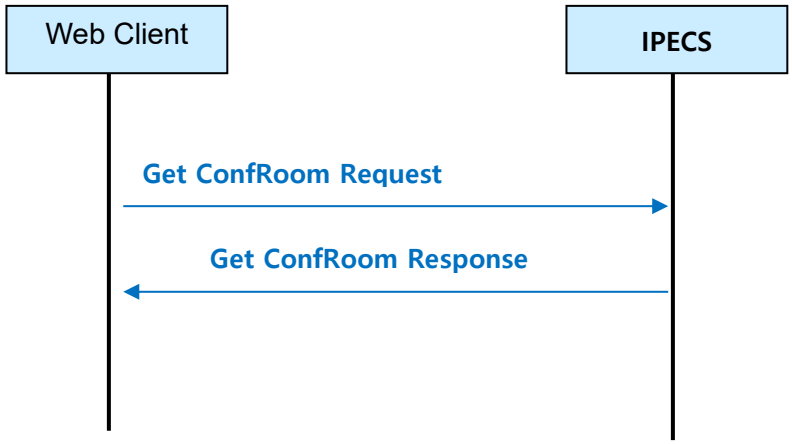
License

API License – CTI required

Usage

Web Client sends Get ConfRoom to iPECS system, and iPECS system issues Positive Response with Conference Room Info, and Negative Response if it is not.

- flow



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/confroom
Method	**GET**
Header	Authorization: Bearer <token from login response>
Parameter	• deviceNumber • roomIndex

Positive Response	HTTP 200 OK
	Content:
	{
	“data”: {
	"roomIndex": 10,
	"roomUsage": “true”,
	"memberCount": 2,
	"member": [“601207”, “601208”]
	}
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	"error" : {
	"code": xxx,
	"message": "xx xxxx"
	}
	}

Request Parameter

Parameter	Value	Mandatory/ Optional	Description
deviceNumber	Digit string	M	Terminal number
roomIndex	Positive integer	M	Room index

Response Parameter

Parameter	Value	Mandatory/ Optional	Description
roomIndex	Positive integer	M	Conference Room Index
roomUsage	string	M	Conference Room Usage true, false
memberCount	Positive integer	M	Member count on Conference Room, Max eMG: 13, UCP: 32
member	[Digit string]	M	Member list on Conference Room

5.3.5.36. Conference Room Open/Close

Function Description

Web Client can open/close Conference Room.

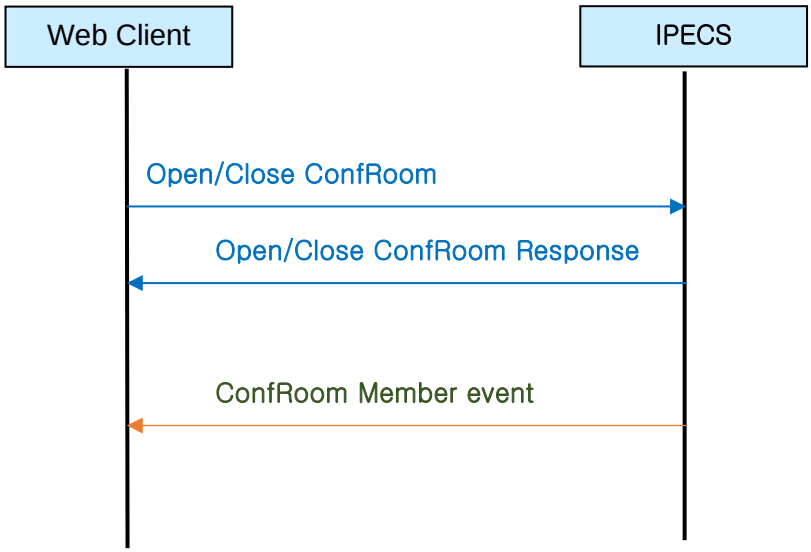
License

API License – CTI required

Usage

Web Client sends Open/Close ConfRoom to iPECS system, and iPECS system issues Positive Response and ConfRoom Member Event if the request command is valid, and Negative Response if it is not.

- flow



URI	{BASE URI}/server/{VERSION}/users/{User ID}/call/confroom
Method	**POST**
Header	Authorization: Bearer <token from login response>
Parameter	• deviceNumber • roomIndex • roomPwd • roomUsage

Positive Response	HTTP 200 OK
	Content:
	{
	“data”: {
	}
	}

Negative Response	HTTP 401 Unauthorized,
	HTTP 403 Forbidden
	HTTP 404 Not Found
	Content:
	{
	“error” : {
	“code”: xxx,
	“message”: “xx xxxx”
	}
	}

Parameter	Value	Mandatory/ Optional	Description
deviceNumber	Digit String	M	Conference Room Member Number

roomIndex	Positive Integer	M	Conference Room Index Range) 1 ~ 9 if “Conference Room Expansion” value is set OFF in “System Attributes(160~161)” menu. 1 ~ 100 if “Conference Room Expansion” value is set ON in case of eMG80 or 100 1 ~ 200 if “Conference Room Expansion” value is set ON in case of eMG800 or UCP
roomPwd	String	M	Conference Room Password
roomUsage	String	M	true or false

5.4. Click to Call

“Click To Call Service” must be enabled in Station Data-> Common Attributes (111).

The user ID is the station number and the password uses Table Data->Station Authentication Code Table(227). ClickCall Application license is required. Only available to station users.

Feature	Clickcall
Description	Click to Call
Service Type	Client
Method	POST
URI	https://{IP ADDR}:{PORT}/ipxapi/client/v1/users/{User ID}/clickcall
Header element	Authorization : bearer {token from system}

Request Data	* User ID is station number in SME system.
	{
	“From” : “{My Station Number}”,
	“To” : “{Destination Number}”
	}

Return Format	Normal case
	200 OK in header without content
	
	Error case
	{
	"error": {
	"code": ###,
	"message": "#Message"
	}
	}

6. Error Code Table

Category	Code	Meaning
Common (0 ~ 99)	1	Parameter error
	2	Version error
	3	License error
	4	Login information mismatch
	99	Unknown
Click to Call (100 ~ 199)	101	Authentication failure
	102	Access denied
	103	Call restricted
	104	Not OK
	105	Parameter error
	106	No access VSF
	107	No VSF group
	108	No access PGM authority
	110	Caller busy
	121	Feature not set
	199	Unknown
SMDR (200 ~ 299)	299	Unknown
CTI	101	generic
	102	serviceNotSupported
	103	invalidMonitorObjectType
	104	invalidCallingDeviceIdentifier
	105	invalidCalledDeviceIdentifier
	106	invalidConnectionIdentifier
	107	invalidDestination
	108	invalidDestinationDeviceObject
	109	invalidDivertingDeviceIdentifier
	110	invalidHeldDeviceIdentifier
	111	invalidParameterValue
	112	invalidForwardingDestination
	140	AssociateBusy
	141	AssociateDND
	142	AssociateFWD

	143	AccessDenied
	144	TimeOut
	145	InvalidAscState
	146	InvalidAscType
	147	InvalidState
	148	InvalidOperation
	149	InvalidPassword
	150	unregistered
	201	NoHeldCall
	202	invalidHeldDeviceState
	203	invalidConnectionState
	204	noCallToAnswer
	205	noConnectionToClear
	301	resourceBusy

Table 5. Error Code Table